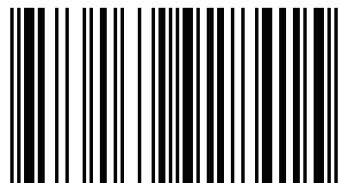


The current practical syllabus of Zoology has been framed Periyar University as well as State level. As there are no such books which cover diagrams only but we have insert the original specimen picture were included. we thought to prepare such a book for students and teachers as well as for those people who have scientific aptitude and interest for the subject. The book is richly illustrated with figures and original photographs. First aid, precautionary methods and safety rules and regulations and also included specific guidelines, how to handling the equipment in the laboratory and also add up the Glossary and Reference.



S. Kalaimani currently working as a assistant Professor in zoology, J.K.K.Nataraja College of Arts and Science, Komarapalayam, earlier he worked at GTN Arts and Science college, Dindugul, and 12 years experience in PG Assistant of Biology in Buds Matric Higher Secondary School. He has diverse experience in Research.



978-613-9-92841-5

Subramani, chithan, Shanmugam

Kalaimani Subramani
Kandeepan chithan
Kowsalya Shanmugam

A Text Book of Practical Invertebrate

LAP **LAMBERT**
Academic Publishing

**Kalaimani Subramani
Kandeepan chithan
Kowsalya Shanmugam**

A Text Book of Practical Invertebrate

FOR AUTHOR USE ONLY

FOR AUTHOR USE ONLY

**Kalaimani Subramani
Kandeepan chithan
Kowsalya Shanmugam**

A Text Book of Practical Invertebrate

FOR AUTHOR USE ONLY

LAP LAMBERT Academic Publishing

Imprint

Any brand names and product names mentioned in this book are subject to trademark, brand or patent protection and are trademarks or registered trademarks of their respective holders. The use of brand names, product names, common names, trade names, product descriptions etc. even without a particular marking in this work is in no way to be construed to mean that such names may be regarded as unrestricted in respect of trademark and brand protection legislation and could thus be used by anyone.

Cover image: www.ingimage.com

Publisher:

LAP LAMBERT Academic Publishing

is a trademark of

International Book Market Service Ltd., member of OmniScriptum Publishing Group

17 Meldrum Street, Beau Bassin 71504, Mauritius

Printed at: see last page

ISBN: 978-613-9-92841-5

Copyright © Kalaimani Subramani, Kandeepan chithan,
Kowsalya Shanmugam

Copyright © 2018 International Book Market Service Ltd., member of
OmniScriptum Publishing Group

FOR AUTHOR USE ONLY

A TEXT BOOK OF PRACTICAL INVERTEBRATE

By

S. KALAIMANI, M.Sc., M.Ed., M.Phil.,

SET (Life Science), SET (Education), DCA.,

S. KOWSALYA, M.Sc., M.Ed., SET

Dr.C.KANDEEPAN, M.Sc., M.Phil., Ph.D.,

FOR AUTHOR USE ONLY

ACKNOWLEDGEMENT

Words find no bounds to express my sincere thanks to all who made this project to come true.

I owe my sincere thanks to **Dr.P.Saraswathi**, Principal, JKK Nataraja College of Arts and Science, Komarapalayam, Namakkal Dt, for extending the facilities to undergo project work. I thank **Dr.Umawathi**, Assistant Professor and Head, PG and Research Department of Zoology, JKK Nataraja College of Arts and Science, Komarapalayam, Namakkal Dt.

My specific thanks to **Dr.S.Umamaheswari**, Assistant Professor, PG and Research Department of Zoology, JKK Nataraja College of Arts and Science, Komarapalayam, Namakkal Dt. for her kind co-operation and encouragement to complete this manual successfully.

I wish to express my profound thanks to **All the Staff Members**, Department of Zoology, JKK Nataraja College of Arts and Science, Komarapalayam, Namakkal Dt.

I also wish to thanks Assistant Professor of the Department of Zoology, My heartfelt thanks to my dear friend **Mrs.S.Kowsalya**, for her cooperation and loving encouragement to complete this manual successfully.

My special thanks to **Dr.C.Kandeepan**, for helping me in many ways for completing this project work.

With great pleasure I express my deep sense of feeling of my affectionate Father **Late Mr.P.Subramani**, Sub Registrar, My Mother **Mrs.S.Veerammal**, My brother **Mr.S.Velumani**, My lovable wife **Mrs.V.N.Padma**, Assistant CTO, My Kids **K.Deepthadhayazhini** and **K.Mithunmonish**, to whom I owe everything for their dedicated efforts and loving encouragement to completed this manual successfully.

And last but not least I wish to extend my sincere thanks to **All My Friends and Relatives** for their co-operation and support in this manual to make success.

S. KALAIMANI

S. KOWSALYA

Dr.C.KANDEEPAN

CONTENTS

Unit No.	Unit Title	Page No.
SPOTTERS		
1	Protozoa : 1.Endamoeba histolytica, 2.Paramecium Caudatum	3 5
2	Porifera: 3.Leucosolenia 4.Hyalonema	9 11
3	Coelentrata: 5.Aurelia Aurelia 6.Obelia colony	15 17
4	Platyhelminthes: 7.Taenia Solium	21
5	Nematoda: 8.Ascaris Lumbricoides	25
6	Annelida: 9.Earthworm 10.Nereis	29 31
7	Arthropoda: 11.Prawn	35
8	Mollusca: 12.Unio	39
9	Enchinodermata: 13.Starfish	43
DRAW LABELED SKETCH		
10	Platyhelminthes: 14.T.S of TaeniaSolium	46

Unit No.	Unit Title	Page No.
	15.T.S of Fasciola hepatica	47
11	Arthropoda: 16.Ephyra Larva 17.Nauplius Larva 18.Zoea Larva	48 49 50
BIOLOGICAL SIGNIFICANCE		
12	Porifera: 19.Sponge Gemmule 20.Sponge Spicules	55 57
13.	Coelentrata: 21.Physalia	59
14.	Annelida: 22.Leech	61
15.	Arthropoda: 23.Limulus 24.Peripatus	63 65
16.	Enchinodermata: 25.Bipinnaria Larva	67
RELATE STRUCTURE AND FUNCTION		
17.	Platyhelminthes: 26.Taenia SoliumScolex	71
18.	Annelida: 27.Earthworm – Body Setae 28.Nereis – Parapodium	73 75
19.	Mollusca:	

Unit No.	Unit Title	Page No.
	29.Penaeus - Petasma	77
20.	Enchinodermata: 30.Starfish – Tube Feet	79
MAJOR PRACTICALS		
21.	Arthropoda: 31.Cockroach – Digestive system 32.Cockroach – Nervous System 33.Cockroach – Male Reproductive System 34.Cockroach – Female Reproductive System	82 83 84 85
22.	Mollusca: 35.Prawn – Nervous System	87
MINOR PRACTICALS		
23.	Arthropoda: 36.Prawn – Appendages 37.Mouth parts of Honey Bee 38.Mouth parts of Mosquito 39.Mouth Parts of Cockroach	89 97 98 99
24.	QUANTA TO MEMORY	101-106
25.	PROCEDURE – GENERAL SAFETY RULE	107-110
26.	SAFETY IN THE BIOLOGY LABORATORY	113-114
27.	SPECIFIC GUIDELINES	117-118
28.	GUIDELINES FOR GENERAL LABORATORY SAFETY	121-124
29.	GENERAL SAFETY RULES IN LABORATORY	127-128

Unit No.	Unit Title	Page No.
30.	EMERGENCY PREPAREDNESS	131
31.	HANDLING EQUIPMENT AND MATERIALS	135
32.	HANDLING BACTERIA AND FUNG	143
33.	CLEAN UP AND DISPOSAL	147
34.	SAFETY PLEDGE	151
35.	GLOSSARY	155-158
36.	REFERENCES	161-162

TEXT BOOK OF PRACTICAL INVERTEBRATES

NATURE AND SCOPE OF BIOLOGY

Biology (GK. Bios – life and logos – to discourse) is the branch of science which deals with the scientific study of life. It is dependent upon other sciences (such as physical and chemical) and thus, has become a multidisciplinary and multidimensional branch of science. Biology is better known as life science', as it seeks to understand the way living beings interact with other living and non-living things.

Precisely, biology is a quest, an ongoing enquiry about the nature of life, Biology, being a major branch of science, requires a scientific approach for understanding and obtaining precise results.

- The term, biology was coined by Lamarck and Treviranus in 1802.
- Aristotle is known as father of biology as well as zoology.
- Theophrastus is known as father of botany.
- Micro organisms were first observed by A.V. Leeuwenhoek but Louis Pasteur is considered as father of Microbiology.
- Emil von Behring (Germany) was the first Nobel Laureate in physiology and medicine who got this award in 1901 for serum therapy against "Diphtheria".

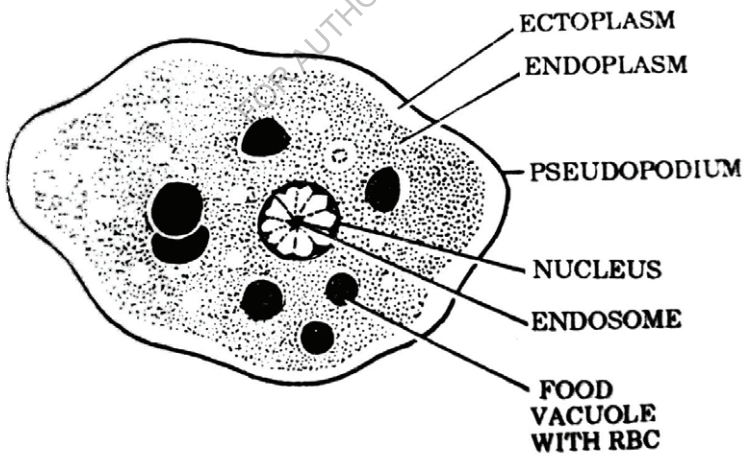
FOR AUTHOR USE ONLY

I. PROTOZOA

INTRODUCTION:-

Protozoa were defined as unicellular protists with animal-like behaviour, such as movement. Protozoa were regarded as the partner group of protists to protophyta, which have plant-like behaviour, e.g. photosynthesis. The term protozoan has become highly problematic due to the introduction of modern ultra structural, biochemical, and genetic techniques, which have showed that the group does not form a clade as required by modern classifications. Modern unicellular clades within Eukaryotes which may be viewed as approximately collectively replacing the class of protozoa include: Excavata, Amoebozoa, Chromalveolata and Rhizaria.

1. ENTAMOEBA HISTOLYTICA



1. ENTAMOEBA HISTOLYTICA

Systematic Position:

Phylum	: Protozoa
Sub Phylum	: Sarcomastigophore
Class	: Rhizopoda
Order	: Lobosa
Genus	: Entamoeba
Species	: Histolytica

Distribution:-

Entamoebahistolytica is worldwide on distribution, but it's prevalence is greater in tropics and subtropics than in the temperate zones. It has been reported that incidence of infection is high in Mexico, China, India and parts of South America.

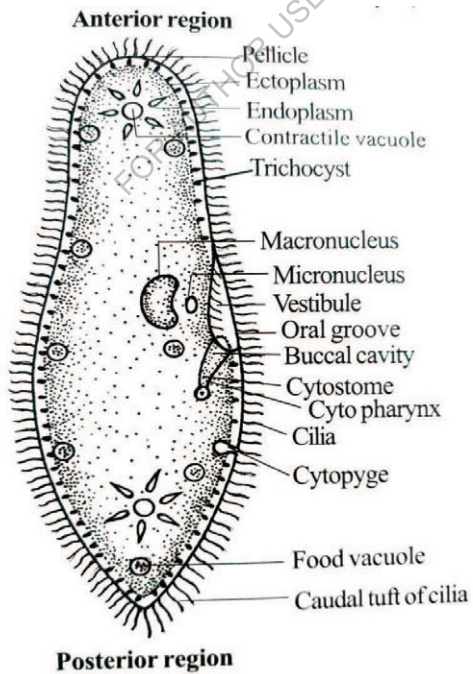
Habits and Habitat:-

Entamoebahistolytica is a microscopic endoparasite of man. It is commonly found in the upper part of the large intestine and is very often lodged in the liver, lungs, brain and testes.

Comment:-

- (i) It is the whole mount of E.histolytica which is a parasite found in the intestine of man and causes amoebic dysentery. In adult trophozoite stage (20 μ m to 40 μ m), the protoplasm has two parts: an outer thick ectoplasm and inner granular endoplasm. Nucleus (4 μ m to 7 μ m) with endosome lies in the centre. It is surrounded by nuclear membrane below which lie the Chromatin granules. Between nuclear membrane and endosome lie network of linin like the spokes of a cycle. Endoplasm contains many ingested RBC, tissue fragments or leucocytes (rarely). The trophozoite moves by blunt pseudopodia.
- (ii) The precystic forms (10 μ m to 15 μ m) are round without any RBC or tissue fragments but has a small blunt pseudopodia. Nucleus and glycogen vacuoles are seen in the endoplasm.
- (iii) Cystic forms may have two or four nuclei. A mature cyst has four nuclei and rod-like chromatid bodies and is transferred with contaminated food or drink (contaminative transference) to a new host.

2. PARAMECIUM



2. PARAMECIUM

Systematic Position:

Phylum	: Protozoa
Sub Phylum	: Ciliophora
Class	: Ciliata
Subclass	: Euciliata
Order	: Holotricha
Suborder	: Trichostomata
Family	: Paramecidae
Genus	: Paramecium

Habit and habitat: Paramecium is best known ciliate, found in fresh water ponds, rivers, lakes, streams and pools, etc.,

Distribution:- It has cosmopolitan distribution

Comments:-

-
1. It is the whole mount of Paramecium which is a free-swimming ciliate found abundantly in stagnant pools and ponds.
 2. Elongated body (0.25 mm) with rounded anterior and bluntly-pointed posterior end.
 3. The entire surface is covered by vibratile cilia.
 4. A wide depression extends obliquely backwards towards the right and ends behind the middle of the body. Peristome leads into the gullet or cytopharynx.
 5. The body is covered by thin pellicle. Protoplasm has an outer clear ectoplasm and inner dense endoplasm.

The ectoplasm contains.

- (a) bascal granules from which arise the protoplasmic fibril which runs in the axix of the cilium and
- (b) trichocysts lying a little deeper in the cortex.

The endoplasm contains

- (a) a large elongated meganucleus,
- (b) a small micro-nucleus
- (c) several food vacuoles and two contractile vacuoles. Contractile vacuoles (Osmoregulation) contract alternately.
- (d) Reproduction by binary fission and conjugation.

Identification:-

Since the animal contains slipper – shaped body and two contractile vacuoles while are star – shaped and has all above features hence it is Paramecium.

FOR AUTHOR USE ONLY

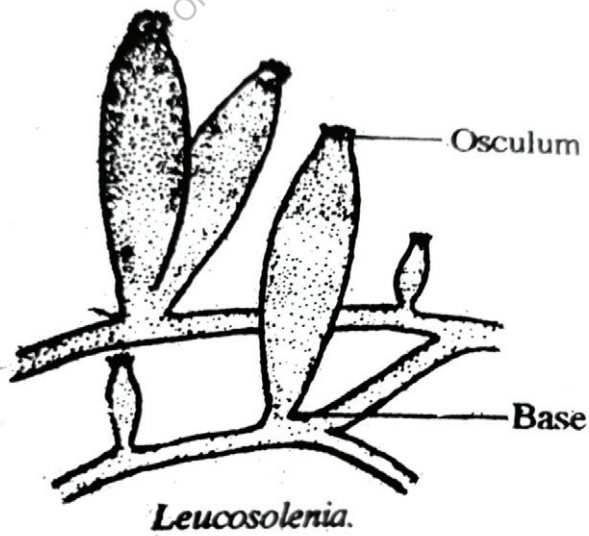
II. PORIFERA

INTRODUCTION:-

Porifera's are multicellular organisms having diploblastic body consisting of outer layer of ectoderm, inner endoderm and an intermediate mesenchyme. Spongocoel is the internal space of the body which is either hollow or permeated by numerous canals lined with choanocytes (collared flagellated cells). Canal system is present which helps in nutrition, respiration, excretion and reproduction in sponges. Mouth is absent and digestion is intracellular. Skeleton consists of spicules (calcareous or siliceous) or fibres (sponging). Asexual reproduction (by buds and gemmules) as well as sexual reproduction (by ova and sperms) takes place in sponges. Fertilization is internal, holoblastic cleavage, indirect development including free-swimming ciliated larval stage called Amphiblastula or Parenchymula. Sponges exhibit great power of regeneration.

PORIFERA

1. LEUCOSOLENIA



PORIFERA

1. LEUCOSOLENIA

Leucosolenia is one of the simplest sponges attached to sea-shore rocks just below the low tide mark. It has the following systematic position:

Systematic Position:

Phylum : Porifera (Pore bearing, diploblastic)
Class : Calcarea (calcareous spicules)
Order : Homocoela (asconid canal)
Genus : Leucosolenia

Comments:-

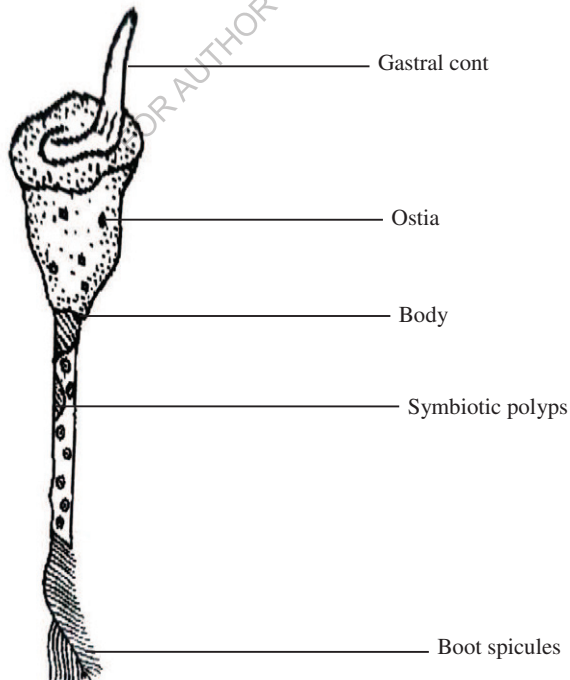
- (i) It is vase-shaped and consists of many horizontal tubes from which branches extend upwards.
- (ii) Each branch has an osculum at its tip and may give branches or buds from the side.
- (iii) Canals originate from the incurrent pores called ostia on the body surface and lead directly into spongocoel. The spongocoel of the branches communicate with each other through the horizontal tubes.
- (iv) Minute organisms and degenerating organic material constitute its food. These are digested in mesenchyme.
- (v) Asexual reproduction by budding and gemmule.
- (vi) Asexual reproduction by Branching and budding.
- (vii) Sexual reproduction by ova and spermatozoa, indirect development including Paranchymula larva.

Identification:-

Presence of osculum, Ostia and colonial structure.

PORIFERA

2. HYALONEMA



2. HYALONEMA

Systematic Position:

Phylum	: Porifera (Porous body, cellular grade of body organization, radially symmetrical)
Class	: Hexactinellida (Skeleton having, six rayed triaxon siliceous spicules)
Order	: Amphidiscophora (Spicules are amphidiscs)
Genus	: Hyalonema

Comments:-

1. Hyalonema is commonly called glass rope sponge as it appears like ball of glass wool with projecting tufts of glassy spicules.
2. It is a marine sponge inhabiting 10-15 metres deep sea water. It is found along the New England coast.
3. Body is round or oval and radially symmetrical.
4. The spicules or root tuft are compact elongated and twisted forming an axis or columella. It helps the organism in anchoring.
5. Several polyps of a Zoanthidea (anemone) grow in symbiotic association with Hyalonema at its columella.
6. Osculum contains a sieve plate.
7. Spongocoel can be noticed only when the upper surface of the sponge body is depressed since the excurrent canals open into it, but when the surface is extended into a gastral cone by upward projection of columella, no spongocoel exists.
8. Skeleton consists of small amphidiscs. Whole body is covered by small branching six rayed spicules which resemble to Christmas trees.

Identification:-

Presence of glossy root spicules tuft, large osculum containing sieve and symbiotic polyps.

FOR AUTHOR USE ONLY

III. COELENTERATA

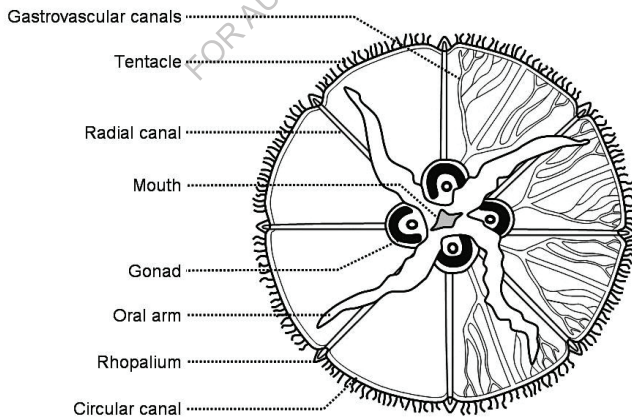
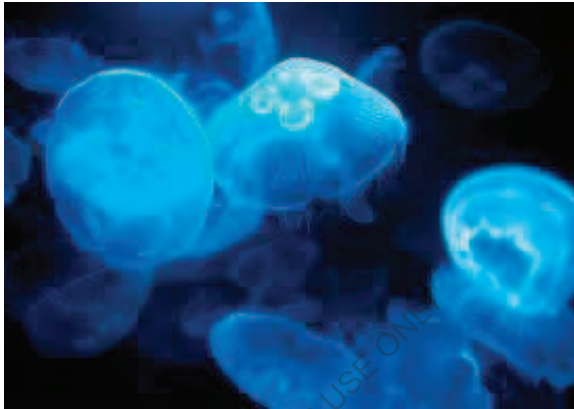
INTRODUCTION

Coelenterata are the Multicellular organisms with tissue grade of organization and are aquatic, mostly marine except a few freshwater forms like Hydra. They are sedentary or free swimming, solitary or colonial and show the radial or biradial with central gastro vascular cavity opening to exterior by mouth. Diploblastic body consisting of outer layer of ectoderm, inner endoderm and an intermediate non cellular gelatinous mesogloea. Acoelomate. i.e. coelom is absent. Mouth is encircled by tentacles bearing nematocysts (stinging cells) meant for food, adhesion, offence and defense.

FOR AUTHOR USE ONLY

III. COELENTERATA

1. AURELIA



1. AURELIA

Systematic Position:

Phylum	: Coelentrata (Tissue grade organization, diploplastic and acoelomate)
Class	: Scyphozoa (Medusa well developed, hydroid reduced)
Order	: Semaestomeae (Free swimming scyphomedusae with umbrella like disc and oral lobes)
Genus	: Aurelia (Jelly fish)

Comments:-

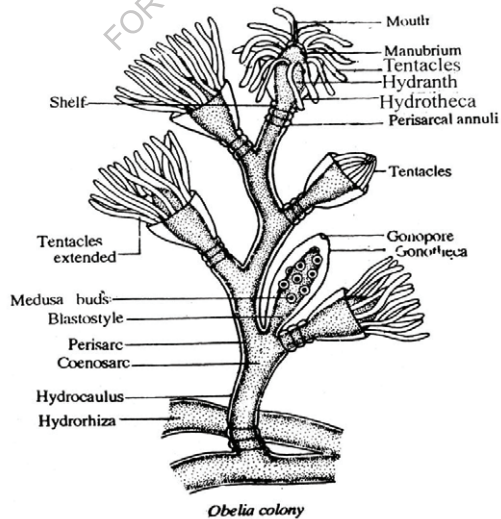
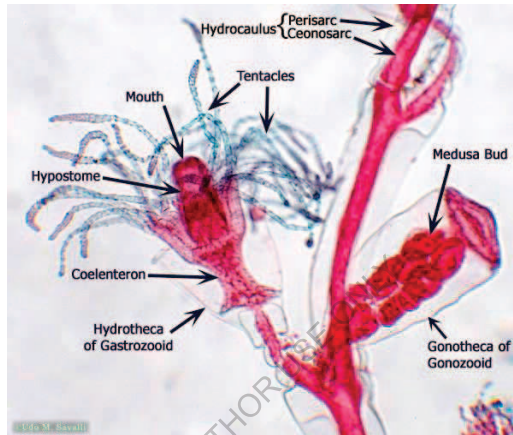
1. Aurelia is commonly known as Jelly-fish. It is a solitary, marine, medusoid form inhabiting warm and temperate seas throughout the world.
2. Body is transparent, bluish white in colour. Inner organs are visible from body surface.
3. Circular body measuring about 90 mm diameter having convex aboral (exumbrellar) and concave oral (subumbrella) surface.
4. Four red or purple horse-shoe shaped gonads are visible from aboral surface.
5. Mouth is four cornered each corner continues into four long and narrow oral lobes hanging down from oral surface.
6. Subumbrellar margin contain marginal tentacles having stinging cells and 8 marginal lappets having sense organs called tentaculocysts (rhopalium)
7. Jelly fish is carnivorous feeding on small animals like molluscs, crustaceans, protozoans, nematodes etc.,
8. It swims by rhythmic contractions of umbrella surface.
9. Aurelia is dioecious i.e. male and female individuals are separate.
10. Life cycle shows alteration of generation.

Identification:-

Jelly like texture, marginal tentacles and oral arms.

COELENTRATA

2. OBELIA



2. OBELIA

Systematic Position:

Phylum	: Coelentrata (Tissue grade organization, diploplastic and acoelomate)
Class	: Hydrozoa (Hydroids, medusa with velum)
Order	: Hydroidea (Polypoid generation well developed)
Genus	: Obelia

Comments:-

1. Obelia is commonly called as sea-fur. It is sedentary marine colonial form attached on the surface of sea weeds, molluscan shells, rocks and wooden piles in shallow waters. It is widely distributed throughout the world.
 2. It is a trimorphic colony present in the form of filamentous sea weed measuring several centimeters in height.
 3. The basal horizontal portion is known as hydrorhiza or stolon. It helps in attachment with the substratum. It gives up vertical branches called hydrocaulus.
 4. Hydrorhiza and hydrocaulus are covered by chitinous perisarc which encloses soft inner coenosarc.
 5. Coenosarc is the living, hollow cellular tube made up of ectoderm, endoderm and mesoglea.
 6. The trimorphic colony of Obelia possesses three types of zooids polyp (hydranths), Gonangium (blastostyle) and medusa (sexual zooid)
 7. Polyp is the nutritive zooid. It is a bell shaped cup made up of lower hydrotheca and upper hypostome. Hypostome carries a ciclet of numerous tentacles provided with nematocysts.
 8. Blastostyle is the club shaped reproductive zooid enclosed within a covering called gonotheca. It gives rise to buds which develop into medusa.
 9. Medusa is the free swimming reproductive zooid consisting of upper exumbrellar (Convex) and lower sub-umbrellar (concave) sides. It is provided with marginal tentacles, four radial canals bearing gonads, a ring canal and a central hanging manubrium on the concave side.
 10. It reproduces by sexual as well as asexual methods. Life history of Obelia exhibits alternation of generation.
-

FOR AUTHOR USE ONLY

IV. PLATYHELMINTHUS

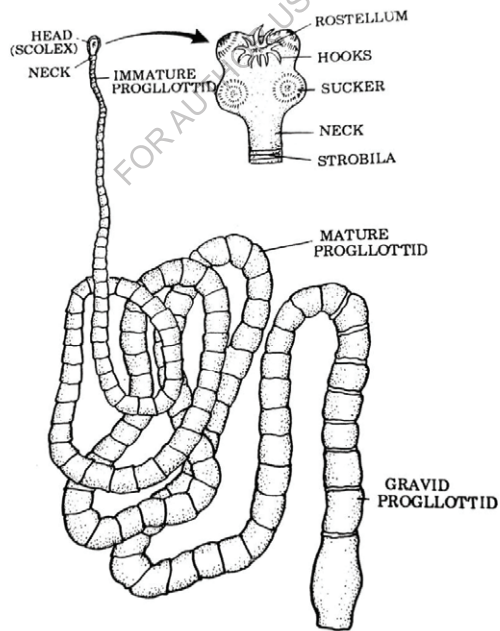
INTRODUCTION:

Platyhelminthus are bilaterally symmetrical and dorso ventrally flattened. Body wall is Triploblastics(form 3 germ layer epidermis,mesodermisandendodermis)below the endodermis longitudinal. Circular and oblique muscles are present.Acoelomate (true coelom is absent)and unsegmented except class cestoda. They are free living,parasite or commensals. Adhesive organ like hooks,suckers and spine are present with adhesivesecretion.

FOR AUTHOR USE ONLY

IV. PLATYHELMINTHUS

TAENIASOLIUM



IV. PLATYHELMINTHUS

1. TAENIASOLIUM

Systematic position:

Phylum	:Platyhelminthes(flatworms,acoelomate)
Class	:Cestoda(endoparasitic,many segments(proglottids)
Subclass	:Eucestoda (ribbon like ,anterior end with an expanded scolex having adhesive structure,each segment each segment with more than one set of reproductive organs)
Order	:Taenioidea:endoparasite,scolex with four suckers
Genus	:Taenia
Species	: Solium

Comments:

-
1. Commonly known as tape worm taeniasolium is found in large intestine of man.
 2. Body comprises of scolex (head), neck and strobili (body)
 3. Scolex is about 1 mm in diameter. In addition to suckers, scolex contain rostellum, which is pigmented and provided with a double row of hooks (number of hooks are about 28 to 32 in number)
 4. Body comprises of large number of segments nearly 800 or more, these segments called proglottid.
 5. Each segment (Proglottid) contain one set of male and female reproductive organs, part of excretory, nervous system and a lateral genital opening.
 6. Life cycle involves two hosts
 - (i) Man-definitive host
 - (ii) Pig-intermediate host
 7. When pigs eat contaminated human faces becomes infacted by bladder worm (Cysticercus larva)
 8. Larva or bladder worm stage doesn't cause any damage to pig.
 9. Human are infected by eating pork (pig meat) and Taeniasolium develops into intestine of adult man.

Identification:-

Contains scolex, neck, great number of segments and all above characters, it is called Taeniasolium.

FOR AUTHOR USE ONLY

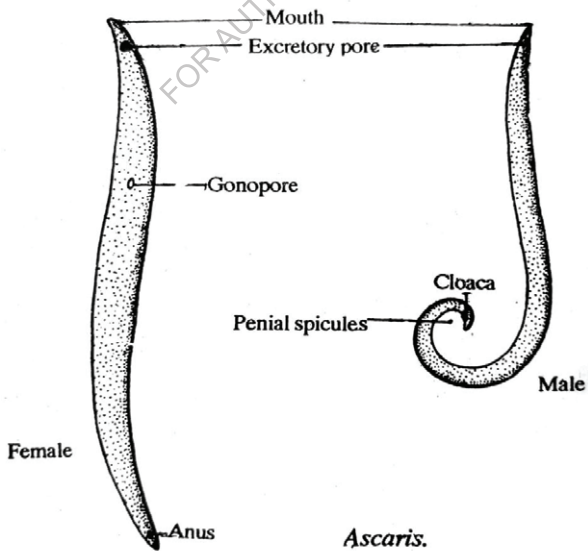
V. ASCHELMINTHES

INTRODUCTION:-

Phylum Aschelminthes (Round worms) inhabit aquatic and terrestrial environment and are well known parasites. They are bilaterally symmetrical and unsegment worms. Triploblastic and pseudocoelomate animals with organ system grade of body organization. Body size mostly small, some microscopic, while others few millimetres to a meters or even more in length. Body mostly cylindrical or flattened, body wall with cuticle and cilia absent. The digestive includes straight and complete alimentary canal with anus, muscular and highly specialized pharynx.

V. ASCHELMINTHES

ASCARIS



V. ASCHELMINTHES

1. ASCARIS

Systematic Position:

Phylum	: Aschelminthes – Unsegmented, Pseudocoelomate
Class	: Nematoda – Cilia absent, round worm, alimentary canal straight
Order	: Living in Interstate of vertebrates, mouth with three lips, pharynx without a posterior bulb.
Genus	: Ascaris
Species	: lumbricoides

Comments:-

-
1. Ascaris is a common endoparasite in the small intestine of the man.
 2. It causes ascariasis in man especially in children. Infection by eating raw & uncooked vegetables.
 3. Shows sexual dimorphism with separate male and female individuals. Male measures 15 to 30 cm and female 20 to 35 cm in length.
 4. Body is elongated, cylindrical pointed at both ends.
 5. Surface of the body is marked with four longitudinal lines.
 6. Mouth provided with a median dorsal and a pair of symmetrical sub median ventral lips.
 7. Excretory pore is small and lies at the ventral surface with distance of 2 mm away from the anterior end.
 8. Tail end of male is ventrally curved containing cloacal aperture, through which two equal isopical projections.
 9. Female genital aperture lies about one-third of the body from the anterior end.
 10. Life history is simple and no intermediate host.

Identification:-

Unsegmented, cylindrical body, posterior end is curved in male, female has pointed tail.

FOR AUTHOR USE ONLY

VI. ANNELIDA

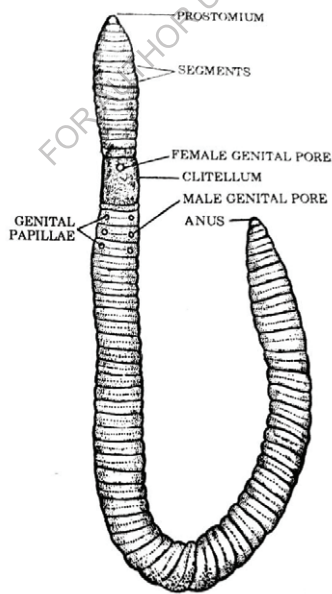
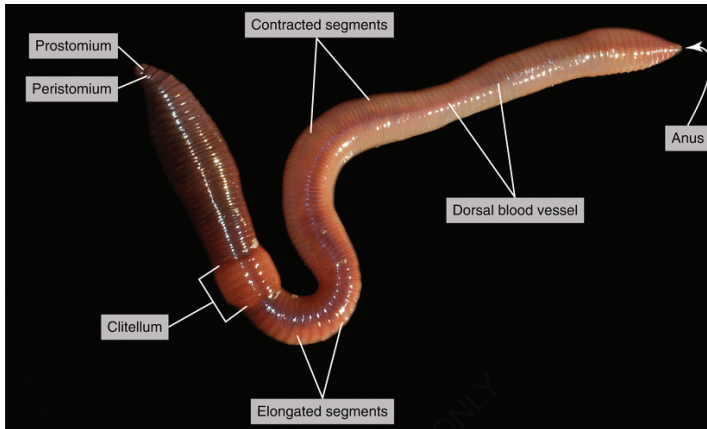
INTRODUCTION:-

Annelida (L., enellus=little rings) known as ringed worms of segmented worms are mostly aquatic, marine or freshwater, some burrowing or tubicolous, terrestrial, free living or sedentary forms. Body elongated, triploblastic, bilaterally symmetrical, metamerically segmented, truly coelomate and vermiform. Body has more than two cell layers, tissues and organs.

FOR AUTHOR USE ONLY

VI. ANNELIDA

1. EARTHWORM



External features of earthworm

VI. ANNELIDA

1. EARTHWORM

Systematic Position:

Phylum	: Annelida
Class	: Oligochaeta
Order	: Neoligochaeta
Genus	: Pheretima

Comments:-

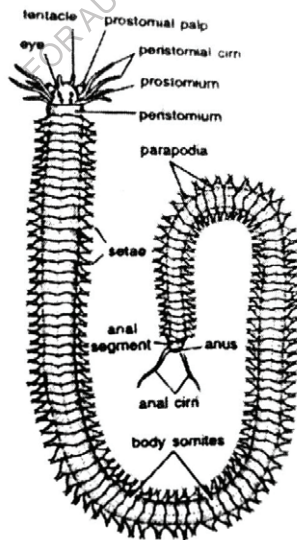
-
1. Pheretima is commonly called earthworm.
 2. Body is bilaterally symmetrical, narrow, long elongated and cylindrical measuring upto 150 mm in length.
 3. Anterior end is tapering and posterior end is blunt.
 4. Body is divided into 100-120 ring-like segments by a distinct series of annular grooves.
 5. Each segment is provided with setae arranged in a ring with each setae arising from a setigerous sac of the skin. The setae help in locomotion by holding the earth.
 6. Mouth is crescentic aperture situated just below the prostomium.
 7. The clitellum is a circular band of glandular tissue which completely surrounds the segments from 14th to 16th segment.
 8. Hermaphroditee.
 9. A pair of male genital pore is situated ventrally in the eighteenth segment while female genital pore are situated at the ventral surface of fourteenth segment.
 10. Anus is situated at the terminal end of the last body segment called anal segment.
 11. Reproduction is usually sexual.
 12. Development takes place in cocoons.
 13. Earthworm is used as bait in fishing and as food by many uncivilized people.
It also has use in medicines, education, experiments and in agriculture as producer of organic fertilizer.

Habit and habita:-Pheretima are burrowers found in the soil rich in decaying organic matters usually in pastures, lawns, gardens, irrigated farm lands, near the banks of ponds, lakes and rivers,

Distribution:-

Found all over world.

2. NEREIS



2. NEREIS

Systematic Position:

Phylum	: Annelida
Class	: Polychaeta
Order	: Errantia
Genus	: Nereis

Comments:-

1. Nereis is commonly called rag worm.
2. The body is long, slender, elongated, dorso-ventrally flattened segmented and is divisible into head, trunk and pygidium.
3. Head consists of two parts, the prostomium and peristomium. Prostomium bears a pair of tentacles, two pairs of eyes and a pair of short two jointed palps. Peristomium bears four tentacles and a slit-like mouth.
4. Trunk is made up of several segments, each bearing a pair of lateral parapodia which are locomotory organs. Setae project beyond the outer margin of each parapodium.
5. Pygidium or anal segment is without parapodia but bears a pair of appendages known as anal cirri and a terminal anus.
6. Respiration happens via blood capillary network of parapodia.
7. Aimentary canal is straight extends from mouth at the anterior end to the anus at the posterior end.
8. Sexes separate. Fertilization is external.
9. The sexual phase of Nereis is known as Heteronereis.

Habit and habital:

Nereis is found in burrows in sand or rocks in intertidal and shallow marine waters.

Distribution:-

Nereis is cosmopolitan in distribution and found in coastal waters of Europe, North Atlantic, Pacific oceans and U.S.A.,

FOR AUTHOR USE ONLY

VII. ARTHROPODA

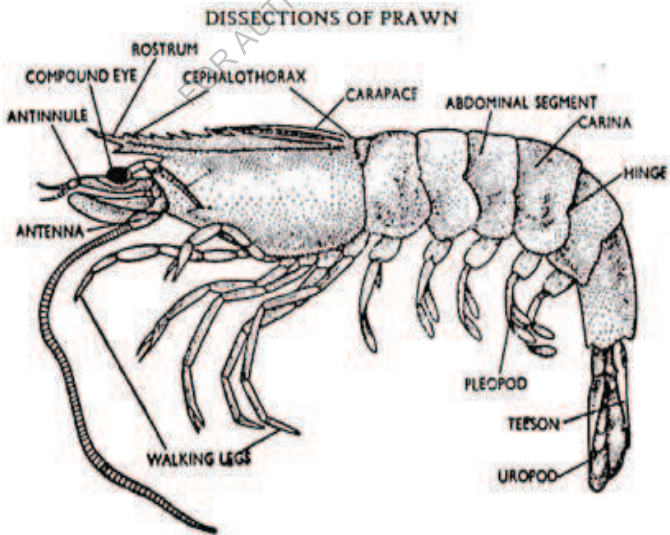
INTRODUCTION

Phylum arthropoda (Gr., arthros = joint + podus = foot) is the largest phylum and most varied in the animal kingdom. Von Siebold gave the name arthropoda. It includes well over one million described species and many million remain unstudied.

Some of the more well-known arthropods include insects, crustaceans, spiders scorpion and centipede as well as the fossil trilobites. Arthropods are mainly terresrial but marie and freshwater species are also well known. Arthropoda varies tremendously in their habitats, life histories and dietary preferences.

VII. ARTHROPODA

1. PALAEMON (PRAWN)



VII. ARTHROPODA

1. PALAEMON (PRAWN)

Systematic Position:

Phylum	: Arthropoda
Class	: Crustacea
Order	: Decapoda
Genus	: Palaemon

Comments:-

1. Palaemon is commonly known as prawn.
2. Body is elongated, spindle shaped and bilaterally symmetrical.
3. Palaemon species are of pale-yellow, pale-blue and greenish color with brown tinge or with orange-red patches. Preserved specimens become deep orange-red.
4. The body can be divided into two regions, anterior cephalothorax and posterior abdomen.
5. Cephalothorax is a large, rigid, unjointed, immovable and cylindrical structure. It consists of 13 segments, 5 of the head region, and 8 of the thorax region.
6. Abdomen is rounded, jointed and compressed laterally. It consists of 6 movable segments and a terminal conical structure, called telson. Each abdominal segment bears a pair of jointed appendages called pleopods or swimmerets.
7. The prawn uses its walking legs for movement at the water-bed.
8. Respiration happens via gills, epipodites and lining of branchiostegites.
9. Excretion through a pair of antennary or green glands, a pair of lateral ducts, an unpaired renal or nephroperitoneal sac and the integument.
10. Sexes are separate. Sexual dimorphism is well developed.

Habit and habitat:-

Palaemon is found in freshwater ponds, lakes, streams, ditches and rivers. It is a nocturnal creature. It hides at the bottom during the day and comes to the surface at night in search of food. It feeds mainly on algae, moss and other aquatic weeds.

Distribution:-

Palaemon is commonly found in India.

FOR AUTHOR USE ONLY

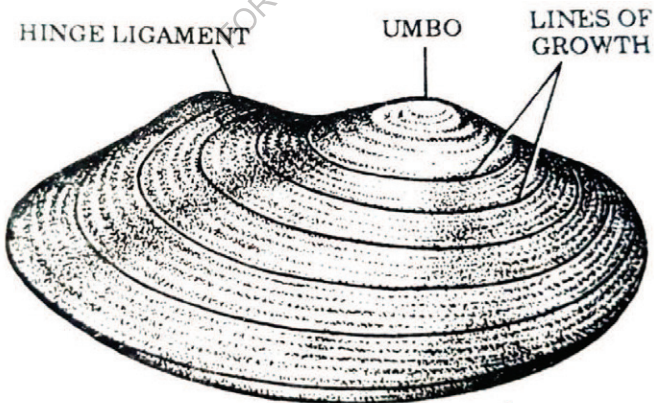
VII. MOLLUSCA

INTRODUCTION

Phylum mollusca(L., molluscus, soft) includes soft-bodied invertebrate animals such as Octopods, snails, slugs, mussels, clams, Oysters, tusk-shells, squids etc. The term mollusca were coined by Johnson. It is a diverse and widespread group, with about 112,000 species. Molluscs are the largest marine phylum, comprising about 23% of all the named organisms. Molluscs may be terrestrial or aquatic (freshwater or marine) They are found in all habitats – deserts, forests, lakes, rivers, abysses of sea, coral reefs, underground or even as parasite in the body of other animals. They may be found clinging to the rocks, crawling, and swimming, burrowing or even digging. They have diversified feeding habit and can be herbivores, carnivores or omnivores. They vary in size from giant squids (nearly 2m long) to little snails, a millimetre long. The largest of the molluscan classes is Gastropoda, which is represented by about 35,000 living and some 15,000 fossil species.

VII. MOLLUSCA

1. UNIO



Unio : lateral view

VII. MOLLUSCA

1. UNIO

Systematic Position:

Phylum : Mollusca
Class : Pelecypoda
Order : Eulamellibranchiata
Genus : Unio

Comments:-

1. There are soft bodied animals characterized by a shell.
2. There are triblastic and coelomate animals.
3. The shell has two valves hence called bivalvia.
4. The foot is wedge shaped and hence the name polychaeta.
5. The gills are basket like.
6. Anterior and posterior adductor muscles are present.

FOR AUTHOR USE ONLY

VIII. ECHINODERMATA

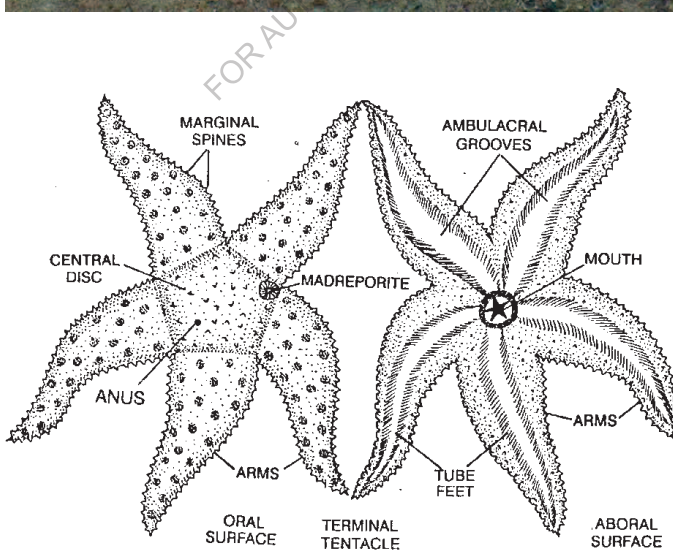
INTRODUCTION:-

Phylum Echinodermata (Gr., echinos=spiny+derma=skin, ata=characterized by) includes exclusively marine invertebrates displaying pentamerous radial symmetry and an endoskeleton of calcareous plates and spines. Jacob Klein gave the name Echinodermata. This phylum is a collection of about 7,000 living species and constitute some of the most beautiful members of sea fauna, such as starfishes or sea stars, sea urchins, sea cucumber, sea lillies and sand dollars etc.,

The name 'starfish' is, however, misleading as these animals are not true fishes. A more suitable name suggested for them is 'sea star'. Common genera of sea stars are Asterias, Pentaceros, Astropecten, Solaster, Heliaster, Luidia, Asterina etc., Genus Asterias includes about 150 species of which most common is A. Rubens

VIII. ECHINODERMATA

ASTERIUS RUBENS (STAR FISH)



1. ASTERIUSRUBENS

(STAR FISH)

Systematic Position:

Phylum : Echinodermata
Class : Asteroidea
Order : Forcipulata
Genus : Asteriusrubens (Star Fish)

Comments:-

-
1. Asterias is commonly known as starfish or sea star.
 2. Average size of Asterias ranges from 10 to 25 cm in diameter.
 3. The body is star shaped, consisting of a central disc with five radiating arms which are broad at their base and tapering towards their extremities.
 4. Mouth is pentagonal and lies in the centre of the disc on oral surface.
 5. Aboral surface bears a number of short, stout spines arranged in irregular rows, numerous dermal branchiae among spines and a smaller aperture, the anus which is situated near the centre.
 6. Pedicellariae are very small, microscopic bodies scattered all over the body.
 7. Sexes are separate. Fertilization is external.
 8. Development includes a free swimming bipinnaria larva.

Habit and habital:-

Asterias are free-living marine animals that can be found at all water depths as well as crawling over rocks and shells Asterias is a marine form and found in the sandy and rocky parts of the sea.

Distribution:-

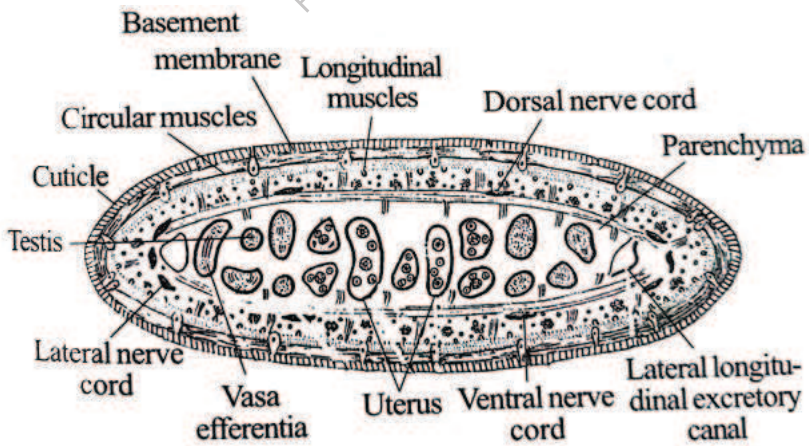
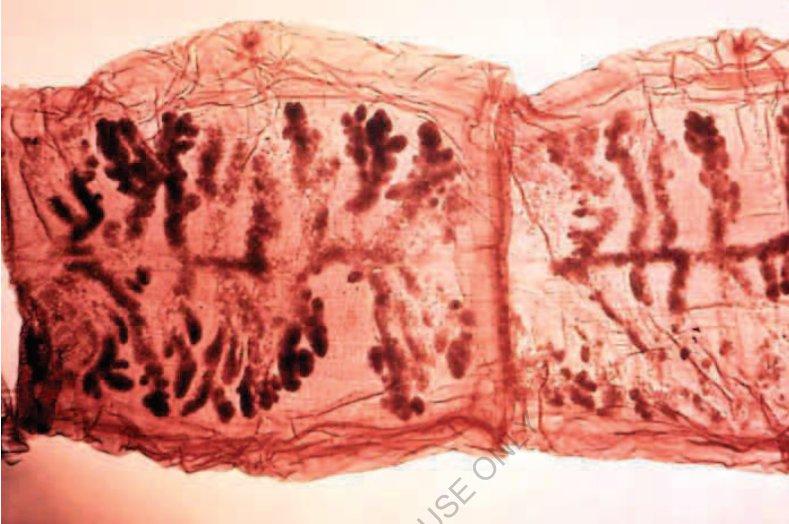
Asterias is found in shallow water in North Temperate seas and found abundantly on North-Atlantic coast. It is found in abundance in India and U.S.A.

FOR AUTHOR USE ONLY

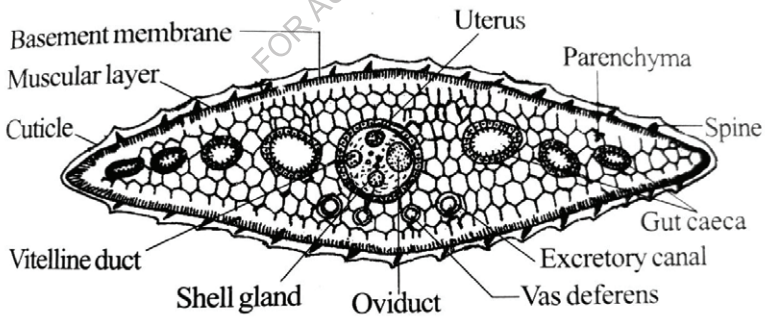
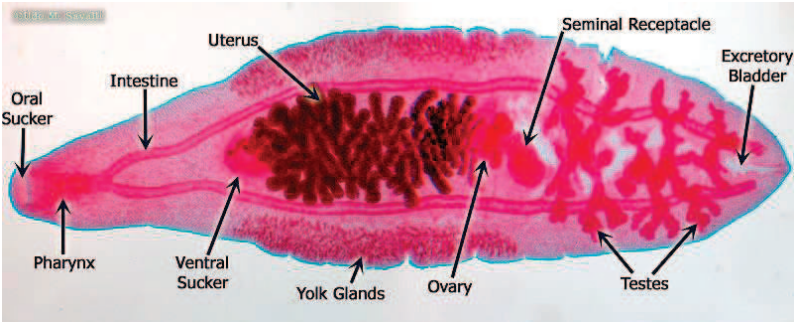
DRAW AND LABEL THE SKETCH

FOR AUTHOR USE ONLY

1. T.S OF TAENIA SOLIUM

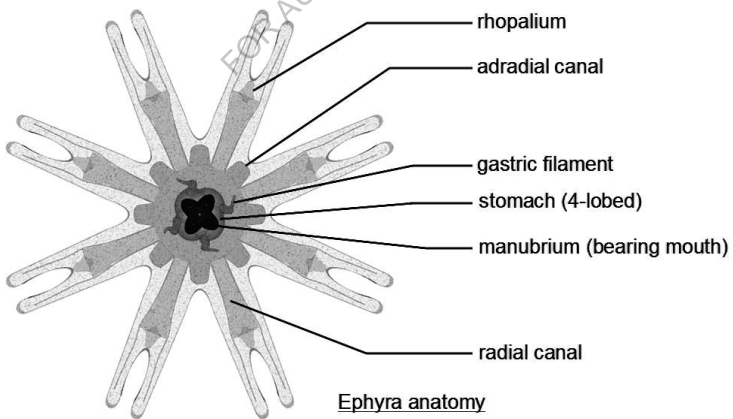
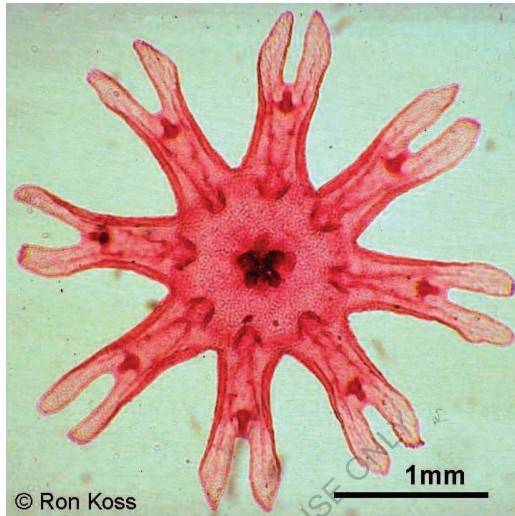


2. T.S FASCIOLA HEPATICA

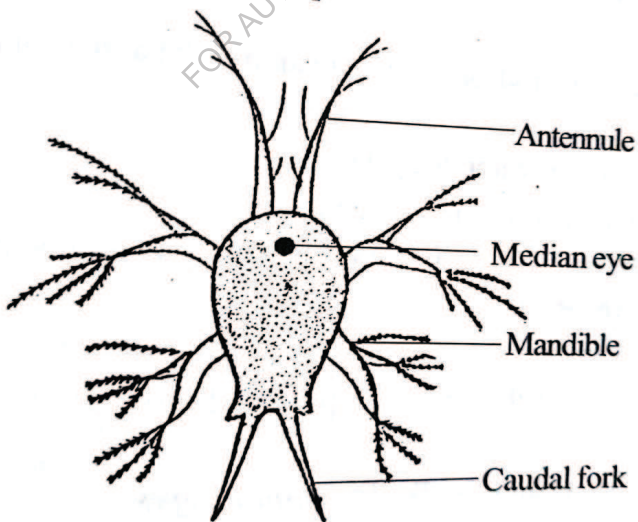


Fasciola hepatica - Transverse section.

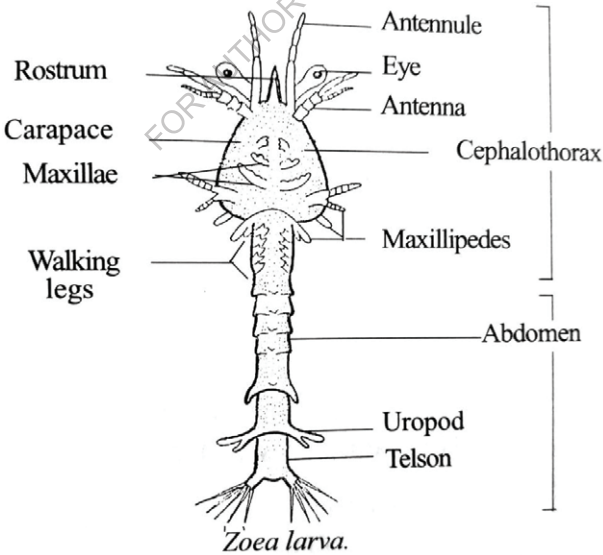
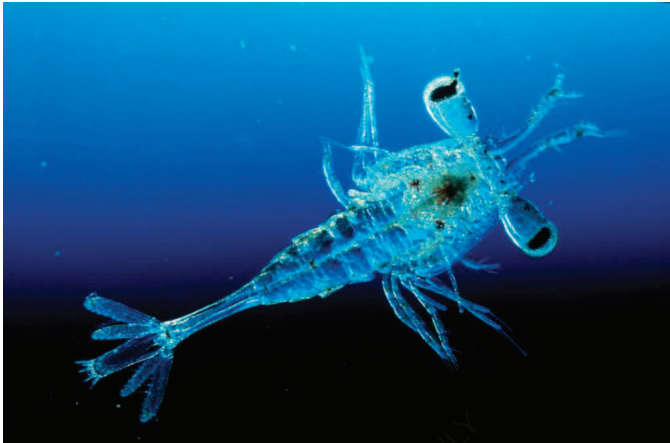
3. EPHYRA LARVA



4. NAUPLIUS LARVA



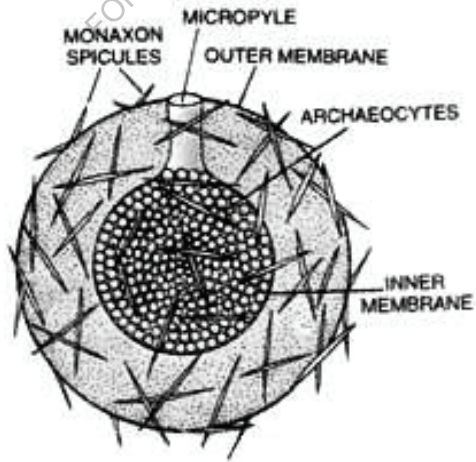
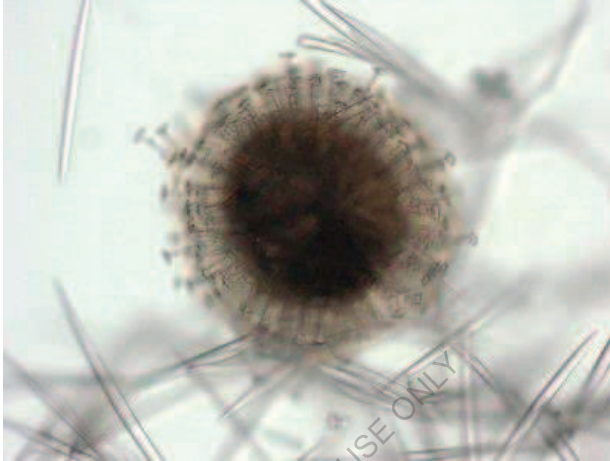
5. ZOEALARVA



BIOLOGICAL SIGNIFICANCE

I. PORIFERA

SPONGE - GEMMULES



I. PORIFERA

1. SPONGE - GEMMULES

Gemmules are asexual reproductive bodies forming a part of regular life cycle. These are endogenous buds which are diagnostics of Porifers, especially of freshwater and a few marine sponges. Gemmulation or endogenous budding is a peculiar mode of reproduction under unfavourable condition such as excessive cold or drought.

Comments:-

Gemmules are asexual reproductive bodies forming a part of regular life-cycle.

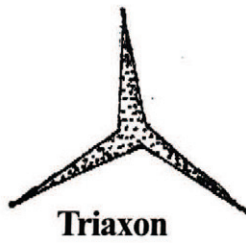
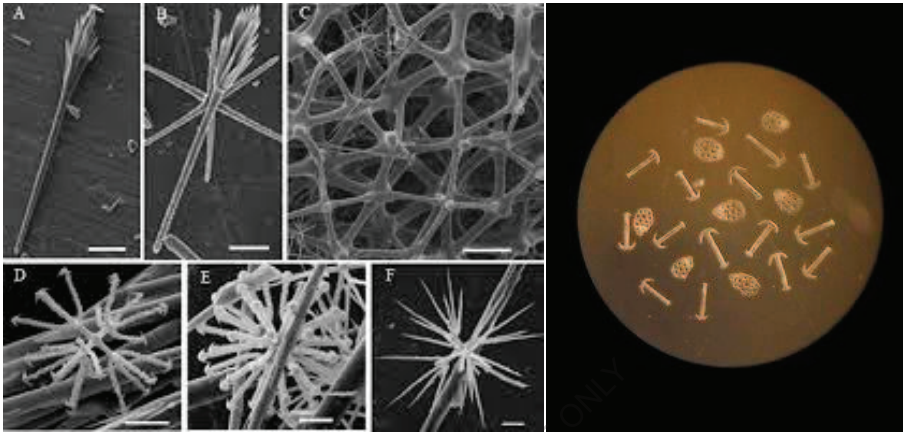
1. Gemmules or endogenous buds are diagnostic of Porifera and especially of fresh-water and a few marine sponges.
2. Gemmulation or endogenous budding is a peculiar mode of reproduction under unfavourable conditions such as excessive cold or draught.
3. Gemmules contain outer and inner membrane.
4. Gemmule is rounded structure formed by the aggregation of archaeocytes into groups accompanied by trophocytes which are impregnated with food particles of glycoproteins or lipoproteins.
5. Scleroblasts secrete the amphidisk spicules, which forms a row in columnar layer between outer and inner membrane.
6. Gemmules are resistant to external factors such as freezing and drying. Gemmules of fresh water sponge can be kept for 2 years.
7. They hatch at a temperature of 13-21 °C in about 3 days. After hatching, a gemmule gives rise to a young sponge.
8. A full grown gemmule is usually pierced by opening on one side, called a micropyle.

Identification:-

Since the material has micropyle in mature and amphidisk spicules in immature gemmules and has above all features, hence it is Gemmule whole mount

I. PORIFERA

SPONGE – SPICULES



Spicules of Olynthus.

I. PORIFERA

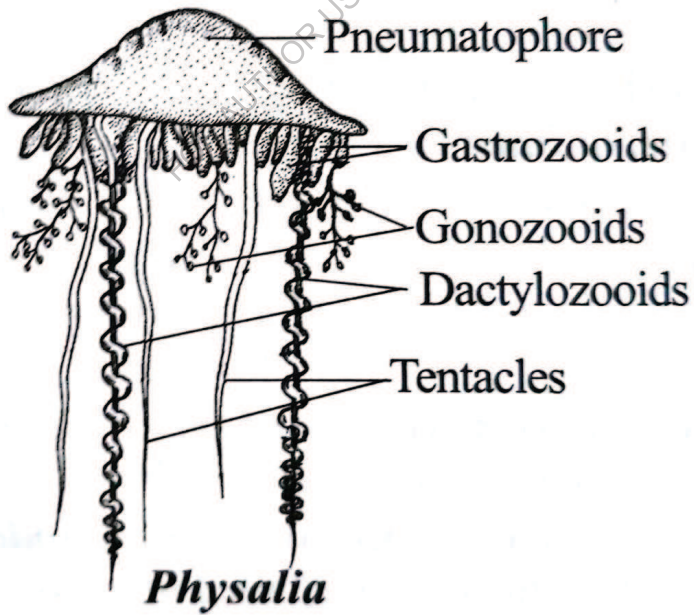
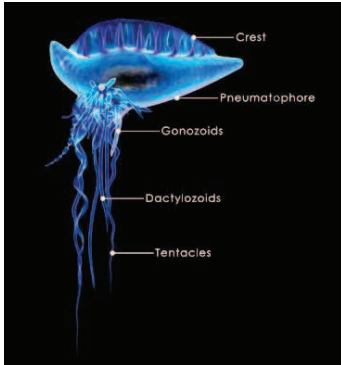
2. SPONGE – SPICULES

Comments:-

1. Sponge body wall is supported by various minute, crystalline and bodies called as spicules, which are secreted by special mesenchymal cells as scleroblasts.
2. Spicules provide taxonomic character and are classified according to the axes and rays, spoken of as axon, actine or actinal respectively.
3. Spicules are of two types. (i) Megascleres or supporting skeleton, (ii) Microscleres small and non supporting. Kinds of Megascleres are as follows;
4. Monaxon consists of a single axis, straight or curved. They may be styles, rhabds and tylots.
5. Tetraaxon consists of four rays. It also includes triradiate or triactinal spicules.
6. Polyaxon having several equal rays. Amphidisk spicules are found in fresh water sponges. In this type, the rhabdom contains disks at both ends. The arrangement of different types of spicules could be seen in Sycon.
7. Microscleres are found throughout the mesenchyme and include spires and asters.
8. Spicules support and protect the body. They are helpful in identification,
9. classification and metabolism.
10. Identification:-
11. The clear transparent monaxon or triaxon spicules indicate Spicule of sponges.

II. COELENTERATE

1. PHYSALIA



II. COELENTERATE

1. PHYSALIA

Physalia is a colonial hydroid commonly called Portuguese man of war. It is a marine, colonial, swimming or floating pelagic animal. /distributed in tropical and subtropical seas.

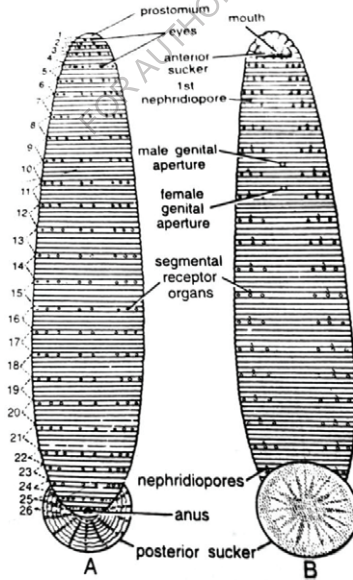
1. Colony consists of large gas filled pneumatophore or float which is bright blue in colour. It floats above the surface.
2. Pneumatophore contains gas glands. It produce a gas having 90% nitrogen, 9% oxygen and 1% argon while swimming the pneumatophore is filled with gas and when animal has to sink the gas is released out.
3. Beneath the float three main types of zooids hang down.
 - (i) Gastrozooids – Nutritive zooids having mouth but tentacles are absent.
 - (ii) Dactylozooids – these help to capture the prey for food. They have numerous nematocyst bearing tentacles.
 - (iii) Gonozooids – Reproductive zooids bearing clusters of male and female medusa.
4. Tentacles are large and bear stinging batteries or nematocysts to kill fishes.
5. The members of physalia colony show division of labour.

Identification:-

Pneumatophore (float), hanging gastrozooids, tentacles and gonozooids.

III. ANNELIDA

1. LEECH



III. ANNELIDA

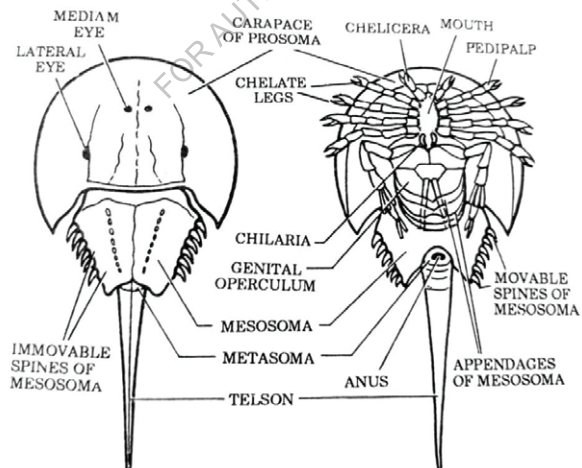
1. LEECH

Hirudinariagranulosa is commonly called Indian cattle leech.

1. Body is soft, elongated, vermiform and bilaterally symmetrical with dorsal surface green and ventral surface orange yellow in colour.
2. During extension of the body is dorso-ventrally flattened while during contraction it is cylindrical.
3. Body of leech is divided into 33 segments. The segments are further divided into annuli or rings.
4. Anterior and posterior suckers are well developed. Anterior sucker is oval bearing triradiate mouth and is formed by the fusion of prostomium with a few somites of anterior region. The posterior sucker is circular in outline and forms a muscular disc at the posterior end. Both the suckers serve powerful organs of adhesion and locomotion.
5. Five pairs of eyes are present on the dorsal side.
6. No special respiratory organs. Respiration happens via skin.
7. Locomotion by creeping/crawling on the surface.
8. Alimentary canal is a straight tube extending from mouth to anus. Small aperture or anus is situated mid-dorsally on the 26th segment at the root of the posterior sucker.
9. Excretory system consists of segmentally arranged seventeen pairs of nephridia arranged from six to twenty-two segments, opening by nephridiopores on the ventral surface.
10. Hermaphrodite.
11. Male genital aperture is situated mid-ventrally in the second and third annuli of 10th segment and female genital aperture between second and third annuli of 11th segment.

IV. ARTHROPODA

1. LIMULUS



Limulus : A. Dorsal and B. Ventral view

IV. ARTHROPODA

1. LIMULUS

Comments:-

1. Limulus is commonly known as king crab.
2. It is bilaterally, triploblastic metamerically segmented animal with jointed appendages.
3. Body is divided into anterior prosoma, posterior opisthosoma and a spine like tail or telson.
4. Prosoma is covered by a horse-shoe shaped carapace. It bears a pair of simple eyes and a pair of compound eyes. It bears 6 pair of appendages surrounding the mouth. The first pair of chelicerac is small, trisegmented and chelate. Rest of the five appendages consists of four pairs are chelate legs and a last pair of non-chelate.
5. Opisthosoma is hexagonal and movably articulated with prosoma. It consists of 6-segmented mesosoma and 3-segmented mentasoma. It bears six pairs of appendages, the first pair is fused and forms the genital operculum and the remaining five pairs of appendages are flap like and membranous and are used for respiration.
6. The telson is used to flip itself over if suck upside down and to steer in the water.
7. Excretion takes place through coxal glands.
8. Sexes are separate. Fertilization is external.

Habit and Habitat:-

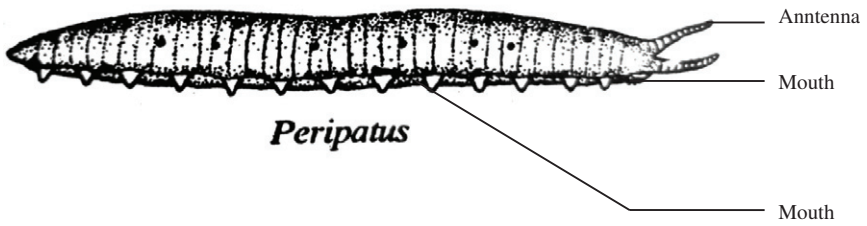
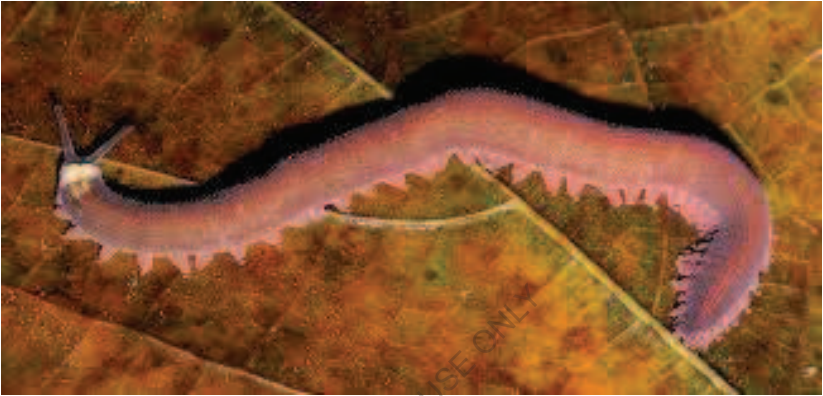
Limulus is marine form and found in shallow waters along sandy and muddy shores of sheltered bays and estuaries. It feeds on molluscs, annelid worms and other benthic invertebrates.

Distribution:-

It is found along the eastern coast line of Asia, Gulf of Mexico and North America .

V. ANNELIDA

PERIPATUS



V. ANNELIDA

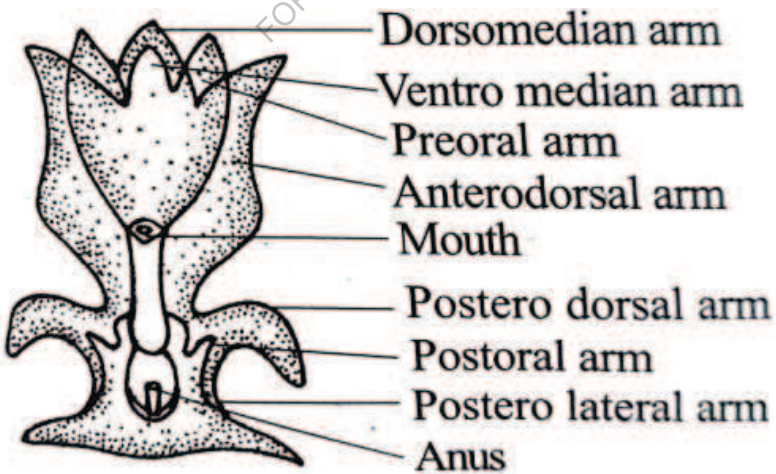
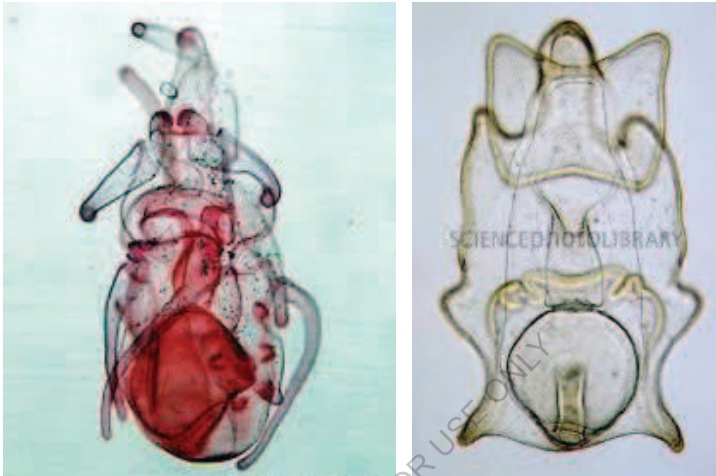
1. PERIPATUS

Comments:-

1. It is found in the forests of Malaya, Australia, New Zealand, Africa, Mexico, Central America and West Indies.
2. It is shy and nocturnal and lives below bark and stanes or holes in rocks, as it likes dark and damp places.
3. The body is cater-pillar like without any external segmentation.
4. Head with a pair of eyes and pair of segmented antennae.
5. The mouth is mid-ventral in position and has a pair of small jaws.
6. Anus at the posterior end and genital pore a little above the anus in between the pair of legs.
7. A series of short segmented legs with claws on the ventrolateral side of the body.
8. The cylindrical body, structure of the eyes, legs, segmentally arranged paired nephridia are Annelidan characters. The Peripatus has some of the Annelidancharacters, on the other hand it has Arthropodan characters like antennae, haemocoel, trachea etc. Thus it represents the living connecting link between the two groups.

VI. ENCHINODERMATE

1.BIPINNARIA LARVA



VI. ENCHINODERMATE

1. BIPINNARIA LARVA

Comments:-

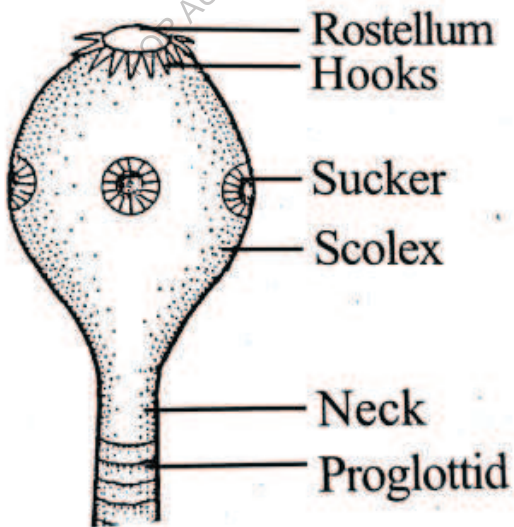
1. This is the whole mount of Bipinnaria larva characteristic of the class Asterozoa.
2. After gastrula the egg develops into this free-swimming ciliated larva.
3. It has one preoral and one postoral ciliary band.
4. The preoral lobe is surrounded by preoral ciliated band.
5. The postoral ciliated band forms a complete ring between the mouth and anus.
6. The prolongations of the body are called arms. They are two posterolateral, two postoral, two posterodorsal, two anterodorsal, two preoral, one ventromedian and one dorsomedian arms.
The anterolateral arms are absent.
7. Alimentary canal is differentiated into mouth, oesophagus stomach, intestines and anus.
8. The larva feeds on diatoms.

FOR AUTHOR USE ONLY

RELATE STRUCTURE AND FUNCTION

I. PLATYHELMINTHUS

1. TAENIASOLIUM – SCOLEX



I. PLATYHELMINTHUS

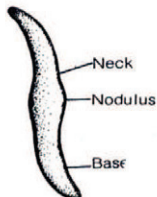
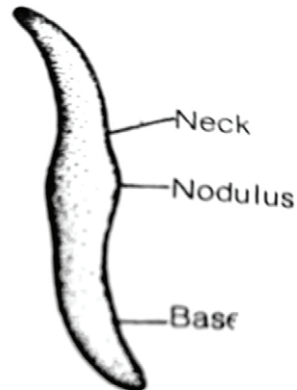
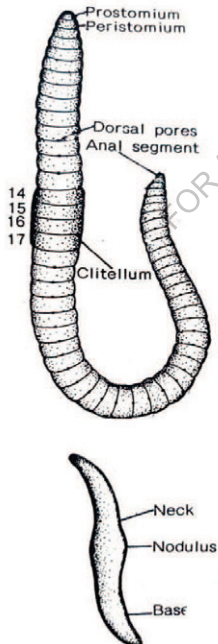
TAENIASOLIUM – SCOLEX

Comments:-

1. In tapeworm the the science is the organ of attachment.
2. It hears **at** its apex a small prominent known as rostellum.
3. Rosetellum is protrussible.
4. At the base of rosetellum, there is a double circlet of eversible chitinous hooks.
5. Four circular unarmed sucker or present.
6. Hooks and suckers serve to fix the animal to the intestine other human host.

II. ANNELIDA

1. EARTHWORM BODYSETAE



II. ANNELIDA

1. EARTHWORM BODYSETAE

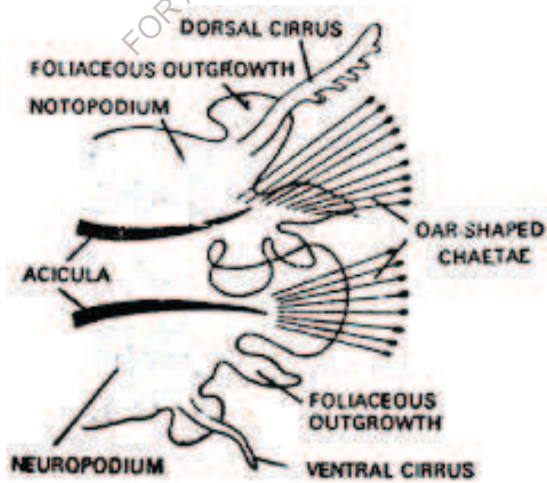
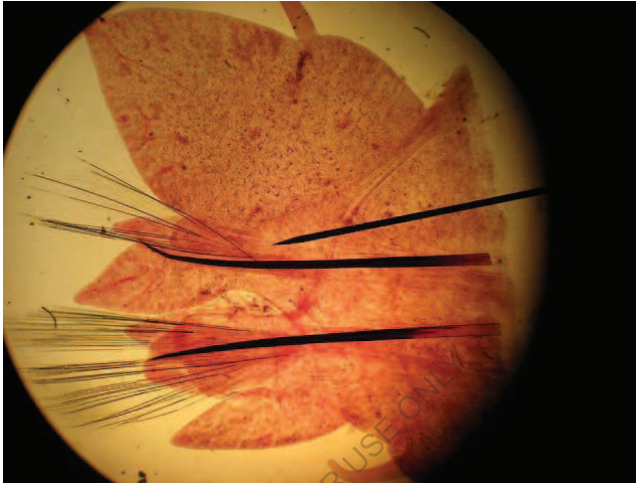
Comments:-

1. The earthworm skin contains numerous bristle like structure is called setae.
2. They are f shaped .
3. They are enclosed in a sac. The setae are formed of chitin.
4. The each setae has three region, namely the nodules, base and neck.
5. The basis located the setal sac.

FOR AUTHOR USE ONLY

III. ARTHROPODA

1. NEREIS-PARAPODIUM



III. ARTHROPODA

1. NEREIS-PARAPODIUM

Comments:-

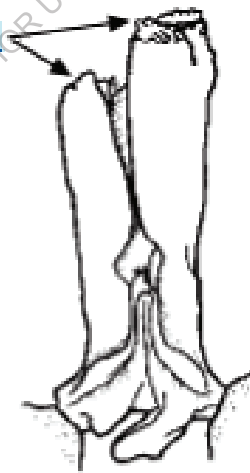
1. Parapodia of Nereis are lateral, muscular, flattened outgrowths of the body wall.
2. The dorsal part bears dorsal circus and bundles of long setae.
3. The ventral part bears a ventral circus and similar bundle of setae.
4. The notopodial and neuropodial setae are embedded in the setigerous sacs.
5. Both notopodium and neuropodium are supported by dark coloured thick chitinous internal rod called aciculums.
6. Parapodia help in respiration and locomotion.

III. ARTHROPODA

2. PENAEUS-PETASMA



asymmetrical



petasma
(ventral view)

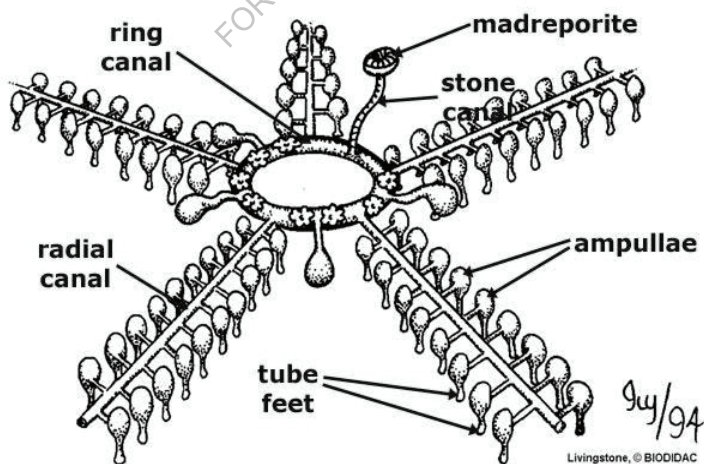
2. PENAEUS-PETASMA

Comments:-

1. The petasma is copulatory organ present only in the male penaeus.
2. It is a membranous structure formed by the fusion of the endopodites of the first abdominal appendages..
3. The male has a pair of tubular testes located in the thorax on the either side of the middle line.
4. The two testes are fused together anteriorly each testes has many finger like outgrowth called caecal diverticula.
5. Posteriorly the testes leads into air vasdeferens .
6. The terminal end of the vasdeferens becomes dilated into a ejaculatory bulb.
7. The ejaculatory bulb opens to the outside by the male genital pore at the base of the last walking legs.

IV. ECHINODERMANTA

1. STAR FISH TUBE FEET



IV. ECHINODERMANTA

1. STAR FISH TUBE FEET

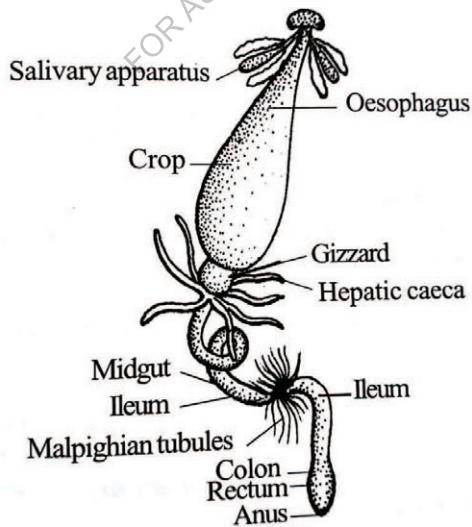
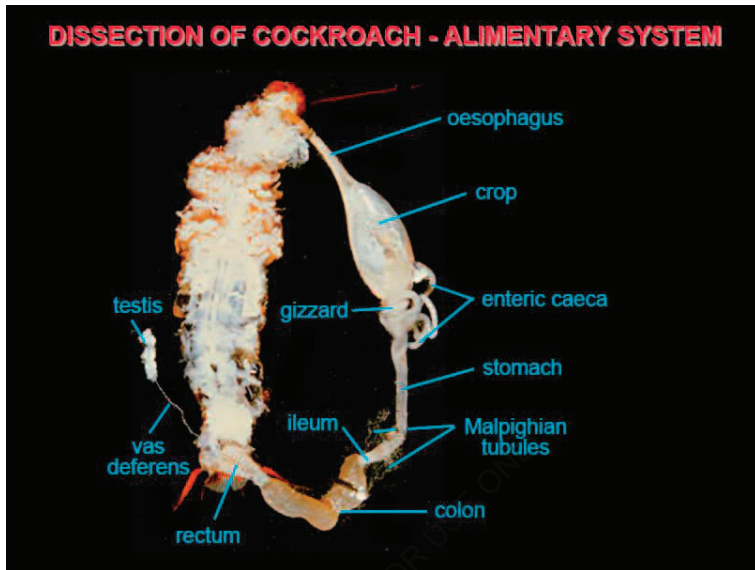
Comments:-

1. The tube feet is a hallow elastic thin walled closed cylinder.
2. It consist of an upper sac like ampulla, middle tubular podium and a terminal disc like sucker.
3. Muscle fibres and present in the walls of the ampulla and the podium.
4. The tube feet has the capable of greated extensions and when extended they come out through the appulateralgroover.
5. The water vascular system has three main function there are us follows.,
 - i) locomotion
 - ii)food capture
 - iii) attachment

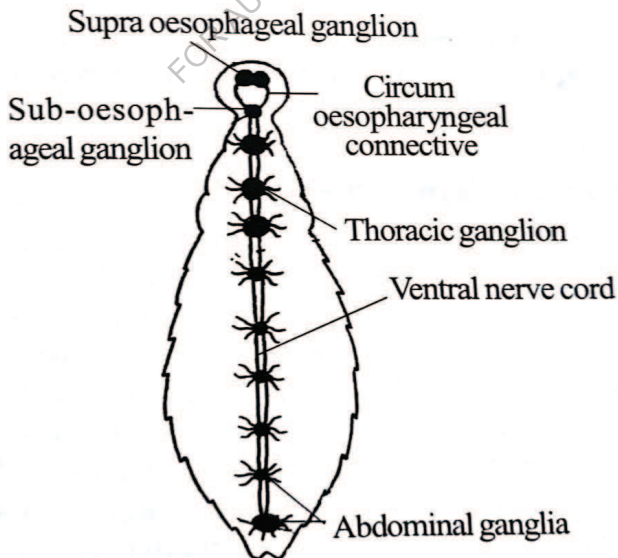
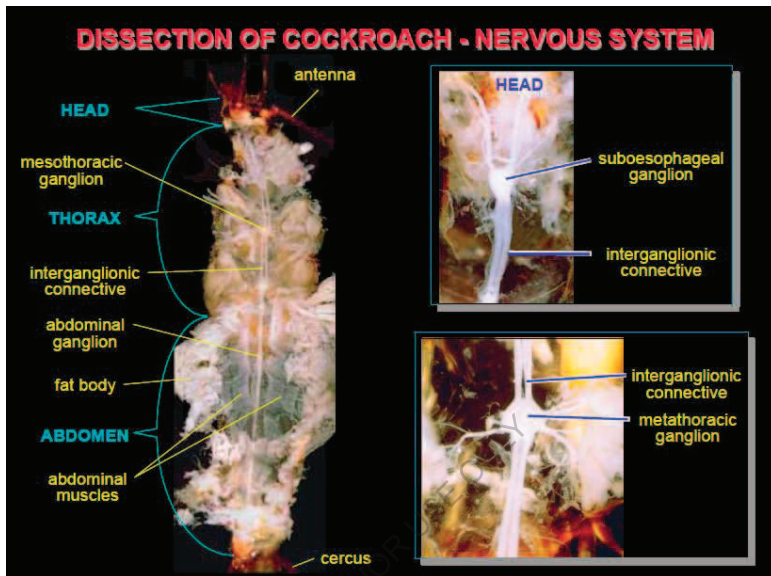
FOR AUTHOR USE ONLY

MAJOR PRACTICALS

COCKROACH – DIGESTIVE SYSTEM

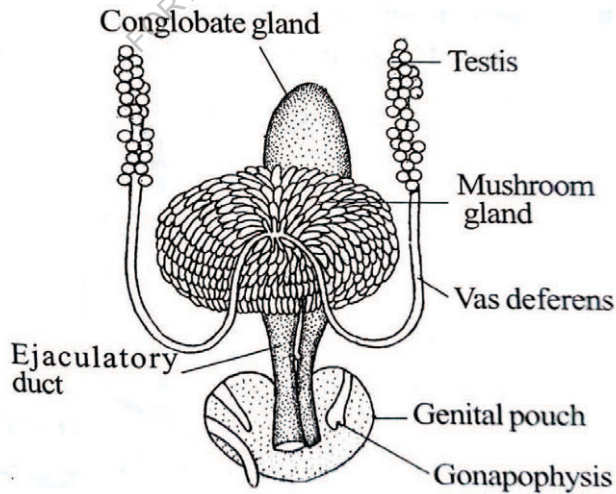
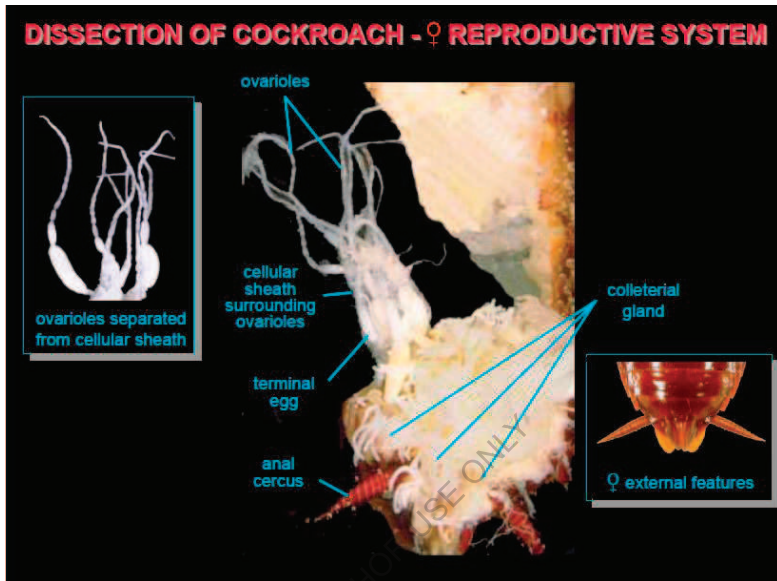


COCKROACH – NERVOUS SYSTEM



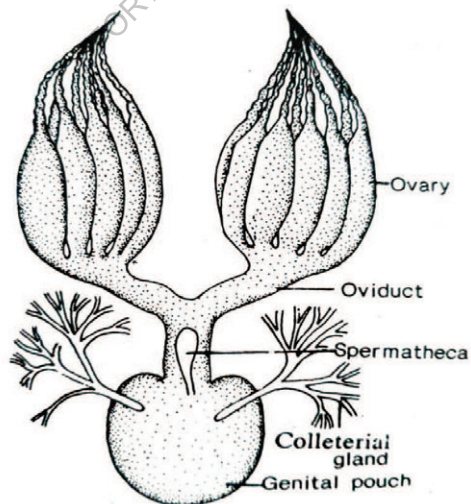
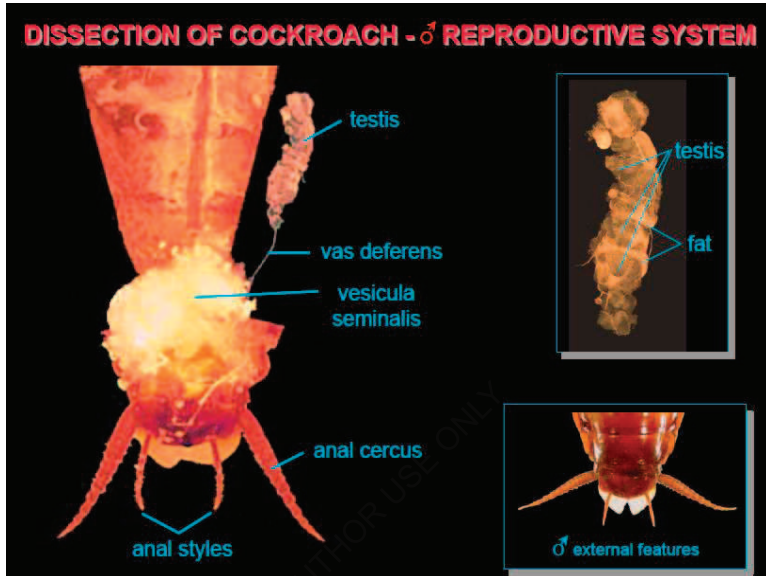
COCKROACH –

MALE REPRODUCTIVE SYSTEM



COCKROACH –

FEMALE REPRODUCTIVE SYSTEM

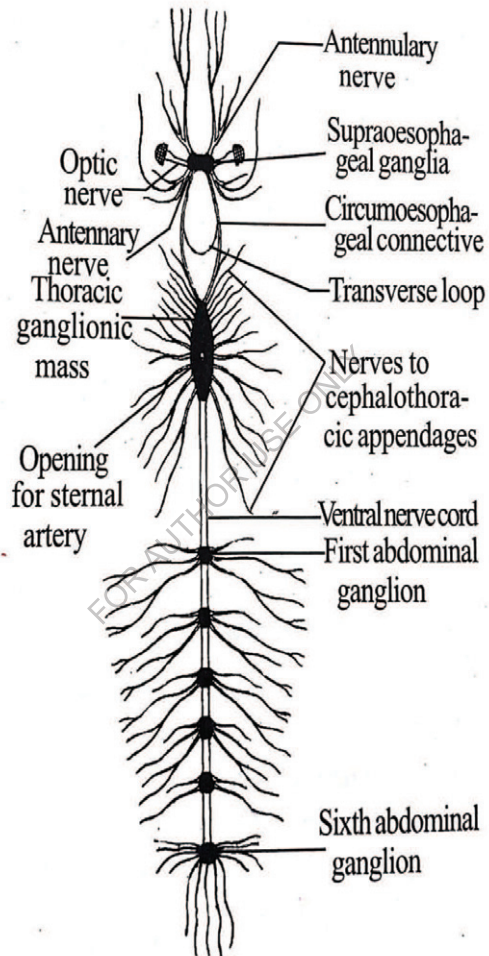


Female reproductive system.

PRAWN – NERVOUS SYSTEM
ORIGINAL DISSECTION



PRAWN – NERVOUS SYSTEM

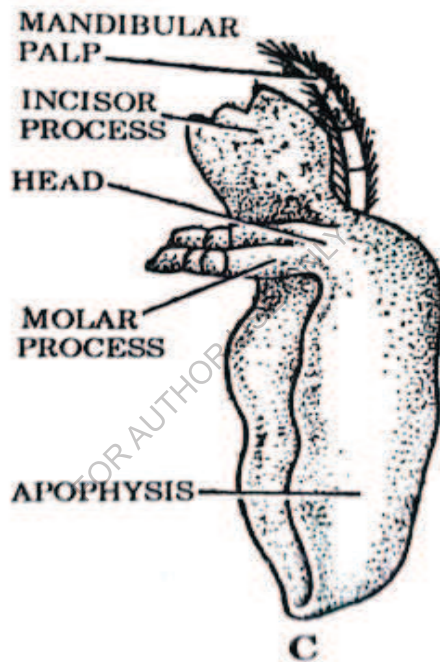




MINOR PRACTICALS

PRAWN – APPENDAGES

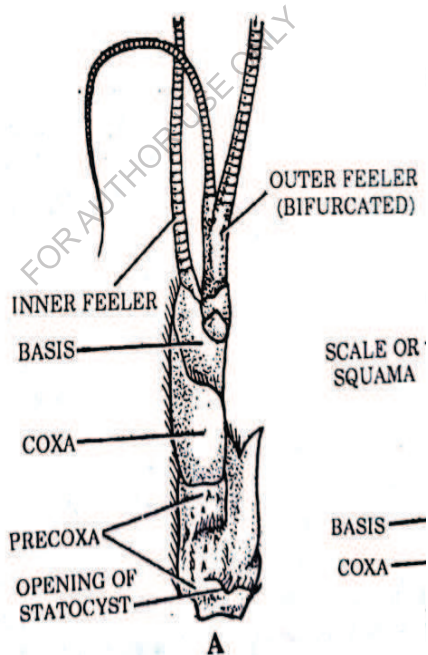
ANTENNULE



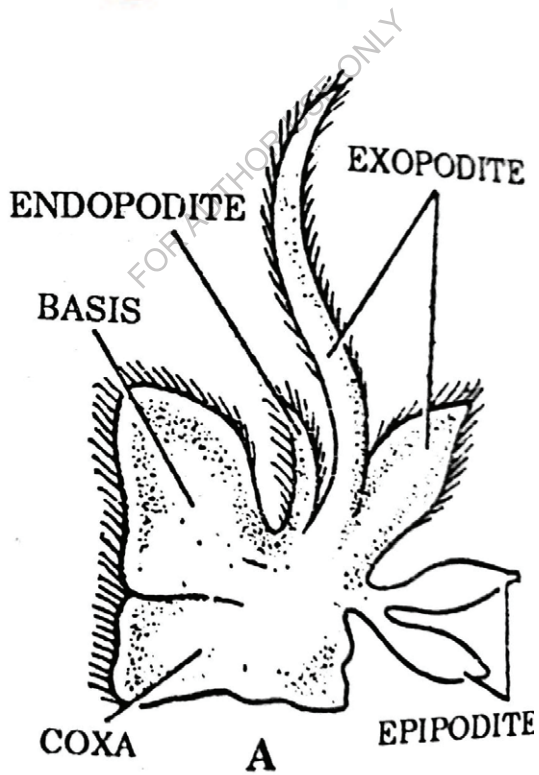
PRAWN – APPENDAGES



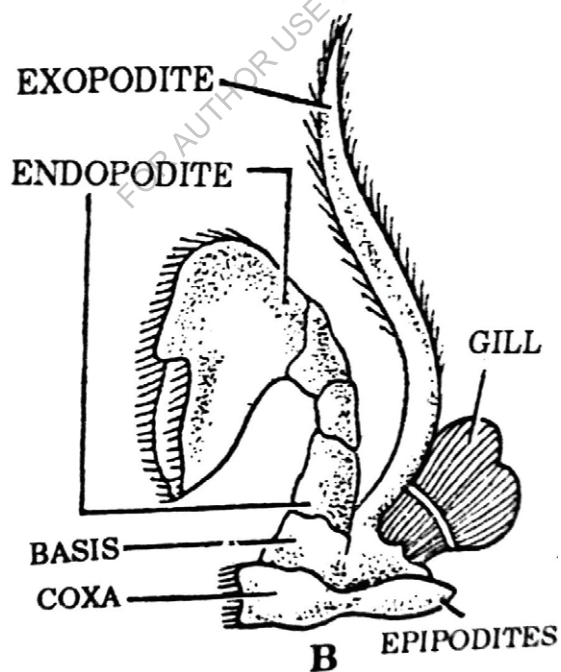
MANDIBLE & ANTENNAE



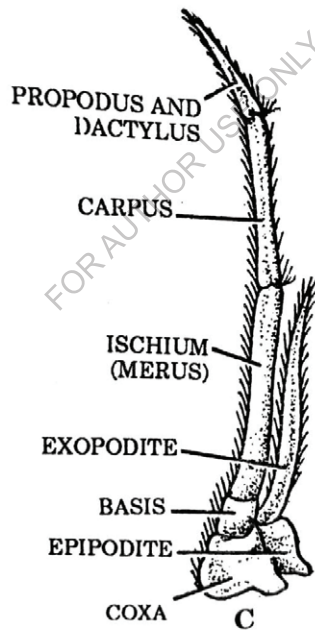
I - MAXILLA



II – MAXILLIPEDES

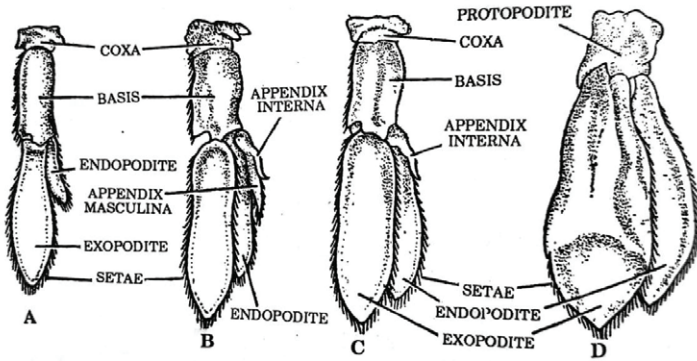


III – MAXILLIPEDES

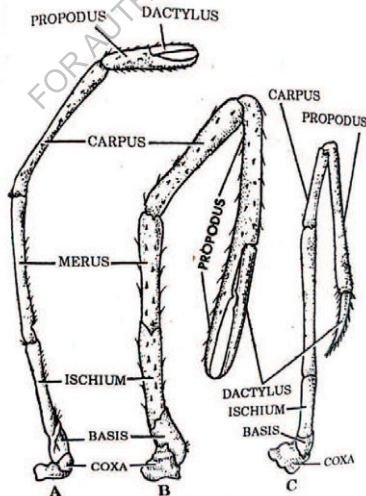


C. third maxilliped

CHELATE LEGS

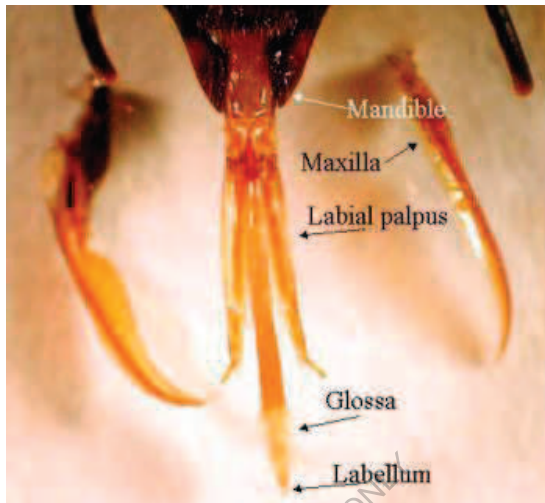


NON CHELATE LEGS

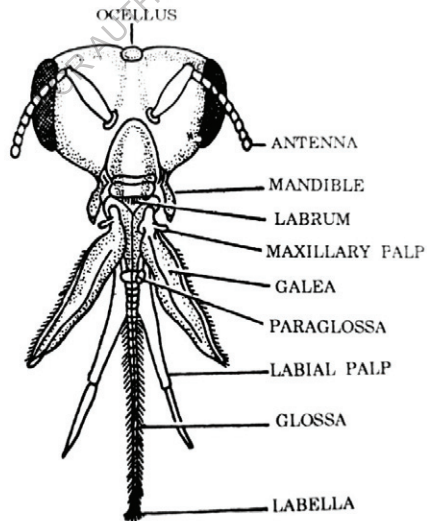


A. first walking leg;
B. second walking leg; C. third walking leg
(3-5 legs have no spincers)

HONEY BEE – MOUTH PARTS

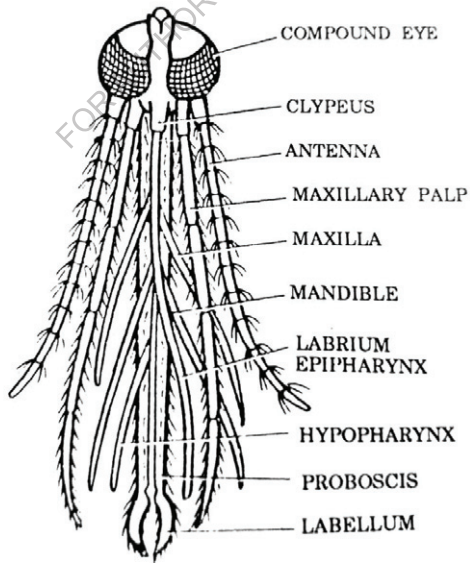
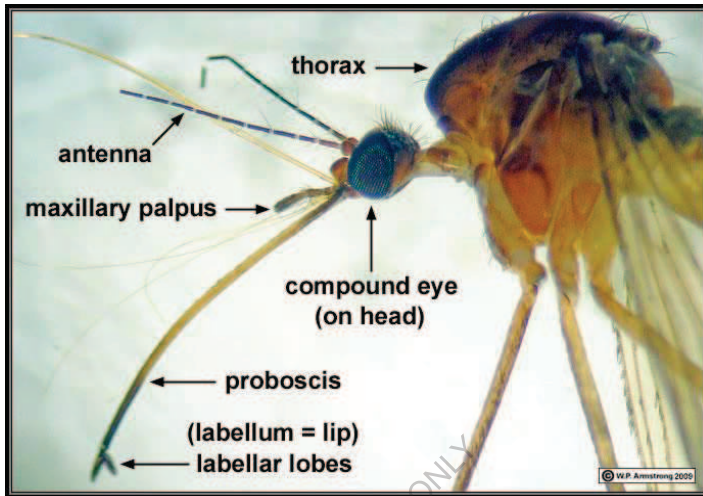


MOUTH-PARTS OF HONEY-BEE (*Apis*)



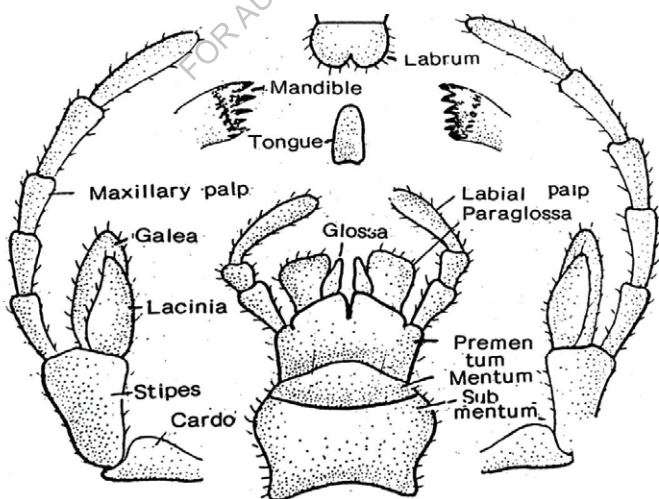
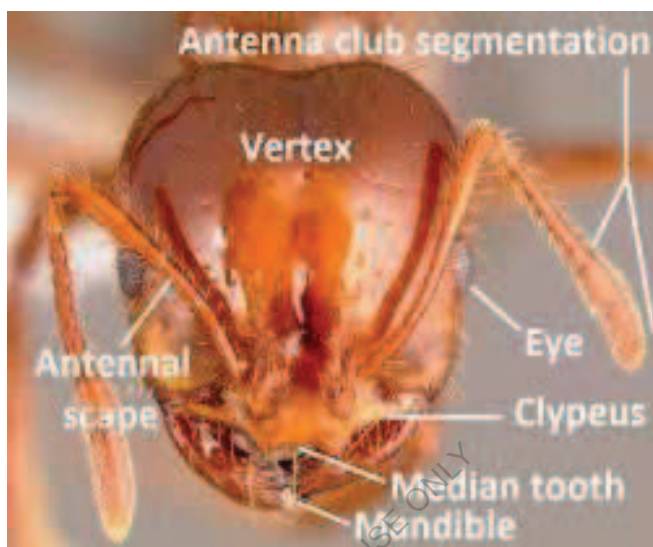
Mouth parts of honey-bee

MOSQUITO - MOUTH PARTS



Mouth parts of female *Anopheles*

COCKROACH - MOUTH PARTS



Cockroach; Mouth-parts.



QUANTA TO MEMORY

- Largest sponge - *Sphaerocapsa*
- Tallest sponge - *Proterion*
- Smallest sponge - *Leucosolenia*
- Some sponges have been found in rocks over 600 million years old.
- Fresh water sponges are found in class *Demospongia*.
- *Spongilla* is a common and widely distributed fresh water sponge.
- *Leucosolenia* is one of the simplest sponges.
- The boring sponges, such as *Cliona* also called as boring sponges, attach themselves to the shells of oysters and mussels. By boring holes into their shells, boring sponges destroy oysters and mussels used as food by them.
- Most probably *Trichoplax* represents the most primitive living metazoans.
- Totipotent interstitial cells are the characteristics of *Hydra*.
- When development starts, *Obelia* is a monomorphic from having polyp only but later due to development of blastostyle it becomes a dimorphic colony and finally medusa but over the blastostyle in a mature colony, then it becomes a trimorphic colony.
- Some cnidarians can paralyse people swimming in the seas. "Sea wasp" a coelenterate is one of the most poisonous animals.
- **Metagenesis:** In the life cycle of *Obelia*, there is regular alternation between fixed polypoid and free swimming medusoid phases, both of which are diploid. Such an alternation between two phases is called metagenesis.

- Some coelenterates which show polymorphism are obelia, Physalia, velella. Physalia shows excessive polymorphism having gastrozooids, dactylozooids and gonozooids.
- In India medusa of Aurelia are commonly found in coastal waters of Madras.
- In the development of Aurelia (a jelly fish) both planula (larva) and ephyra (larva) are present.
- In hydra, only so tissue equivalent to connective tissue of higher animals is found.
- In hydra smooth muscles are present. Female is about 5-12 cm. Male is about 1.5 to 3.3 mm. Male lives in the female's body.
- Mouth of Ascaris is triradiate.
- Ascaris remains attached to intestine as host by its muscular lips.
- Ascaris is both ammonotelic to and ureotelic.
- Male Ascaris is monorhic. It has a single testis. Female Ascaris is didelphic. It has two ovaries.
- First stage juvenile of Ascaris is called rhabditoid larva. Circular muscles are absent in ascaris.
- Female Ascaris lays 20000 to 205 lakh eggs per day.
- Most common parasitic worm today man is Enterobiasvermicularis.
- Oil of chenopodium is used for treatment of Ascaris infection.
- Largest Annelid-Microchaetus spp.
- Smallest Annelid-Chaetogasteranandale.
- Most primitive annelid-Polygordius.
- An earthworm can make its body nearly stiff by means of what is known as its "hydraulic skeleton". The skeleton is its coelomic fluid exerting pressure over the body wall.
- Hirudin is costly anticoagulant used treatments of various diseases.

- Dr.K.N.Bahl of Lucknow University described Pheretimaposthuma in a memoir.
- Chloragogen cells or yellow cells of earthworm are analogous to vertebrate liver cells.
- Brown colour of earthworm is due to pigment porphyrin present in body from bright sunlight.
- Neurohormones are secreted by cerebral ganglia for control of gametogenesis and regeneration.
- Success of arthropod is attributed to chitinous exoskeleton.
- Almost all crustaceans live in water. Those which live on land. Like wood lice (Oniscus) can exist only in damp places. The word 'crustacea' means 'Crusty ones'. This is a suitable name for a group of animals which all have a jointed crust or hard covering to support and protect them.
- Large glutinants are also known as streptolineglutinants.
- The coral reefs are three kinds. They lie close to the shores of some volcanic island or part of some continent are called fringing reefs. They are located some distance from the shore. The stretch of water or lagoon separates the barrier reef from land. An atoll reef is also called coral island or lagoon island, is a circular or horse shoe shaped reef that encircles a lagoon but not an island.
- Lakshadeep is fortified by the coral reefs.
- Coral building polyps grow in seas where temperature of water never falls below 20°C.
- Tapeworm is hermaphrodite, endoparasite and remains attached to host with the help of suckers and hooks.
- Schistosoma is blood fluke. Its primary host is man and secondary host is snail.

- Tapeworm are said to be “Wallowers” because they absorb nourishment through their body surface.
- Life history of *Fasciola hepatica* includes 5 larval stages: Miracidium, Sporozoite, Redia, Cercaria and Metacercaria.
- Largest flat worm is *Taenia solium* is about a few metres long.
- Tegument is the name given to the body covering in flat worms.
- *Microfilaria* and *Filaria* live in lymph glands but microfilariae appear in peripheral blood circulation at night.
- Liver fluke and tapeworm are digenetic. Have two hosts. *Ascaris* is monogenetic has a single host.
- *Ancylostomadum duodenale* (Hookworm) was discovered in 1838 in an autopsy of an Italian peasant woman by Dabini.
- *Bonellia viridis*. In this annelid, sexual dimorphism is very remarkable.
- Usually the larger an animal is, the longer it takes to grow up. One exception is the 17 year cicada (an insect), which spends a very long time as a larva.
- In spiders, web is secreted by last body segment.
- The thread of spider, web is stronger than steel thread.
- The endoskeleton is present in the head and thorax of the cockroach. Abdomen has no endoskeleton.
- In all insects the intervals between two ecdysis (casting of skin) is known as the stadium and the form assumed by the young insect during a stadium is called an Instar.
- Mites are parasites of animals as well as plants.
- Fireflies or glow worms bear light producing organs upon their abdomen. Light is emitted at short intervals by both male and female as a visual signal to attract each other for copulation.

- Gambusia is a larvivorous fish which feed up larvae of mosquitoes.
- It is a Largest mollusc 55 feet (Giant squid) Largest Invertebrate also.
- Largest tentacles or arms - 35 feet (Octopus)
- Largest shell - Tridachnaderesa (4 feet)
- Smallest shell - Ammonicera rota 0.02 inch
- Largest snail - Tethys
- Largest eye of mollusc - Architeuthis (Giant squid)–15” diameter
- Slowest moving snail - Helix (Garden snail)
- Largest Octopus - Octopus apoliyan
- Largest gastropod - Hemifuscus (18”)
- Slowest growth in mollusc - Deep Sea-clam.
- Lac is secreted by female lac insect.
Fastest Flying Insect - Cephemyiapratii
- (818 miles/hr)
- Loudest Insect – Cicada (its sound can be heard from a distance of even 4 kms.
- Largest Grass - Sillicoferggrandis
- Hopper - its wings stretch is about 30 cms)
- Largest ant - Dinoponeragrandis (3.25 mm)
- Smallest Crustaces – Allonella (0.025)
- Heaviest CrustaceaHe – Psedocarcinusgigas
- Largest Crab - Macrocheriakaempfera
- Most Private - Peripatus Arthropod
- Heaviest Insect - Goliathusgoliathus (Beetle)
- Poisonous spider - Lectodetectusmeactans
- Smallest spider - Patumarplesi
- Largest spider - Theraphosa sp.

- Larvae of genus Chironomus (an insect) are called 'Blood worms'. They have red blood. The red colour of these larvae is due to haemoglobin.
- In coastal areas like Konkan Maharashtra of India, the red and black species of scorpions are deadly poisonous. Their venom is fatal.
- Honey is flower nectar mixed with saliva of bees and is natural antiseptic.
- Bees communicate by complex dance.
- Royal Jelly – a substance secreted by honey bee workers.
- Termites are social and polymorphic insects.
- Termites form large nests called termatorium.

FOR AUTHOR USE ONLY

PROCEDURE: GENERAL SAFETY RULES

I. RESPONSIBILITY

The laboratory instructor and laboratory aides are responsible for maintaining this procedure.

II. PROCEDURE

Laboratory Procedures All employees working within the District laboratories are provided with a copy of applicable Laboratory Safety System procedures (chemical Hygiene Plan) and provided with training necessary for safe operations. Work study students are also provided with safety training. All employees are required to familiarize themselves with the procedures and sign a statement acknowledging receipt and understanding of the procedures. The signed statement is kept on file by the Chemical Hygiene Officer.

1. Laboratories are used for laboratory activities ONLY.
2. If you make a mess, you are responsible for ensuring it is cleaned up.
3. Disinfectant is used to wipe lab working areas before and after lab activity.
4. Good laboratory practices are to be followed at all times. If a student is in doubt, direction of the instructor or laboratory aide should be sought before proceeding in ANY activity.
5. All laboratory assignments MUST take place and be completed in the laboratory. NO lab activity is allowed in classrooms.

ALL LABS MUST BE COMPLETED ON THE ASSIGNED DATE. NO OPEN LABS ARE PERMISSABLE UNLESS AN INSTRUCTOR OR LAB AIDE IS PRESENT.

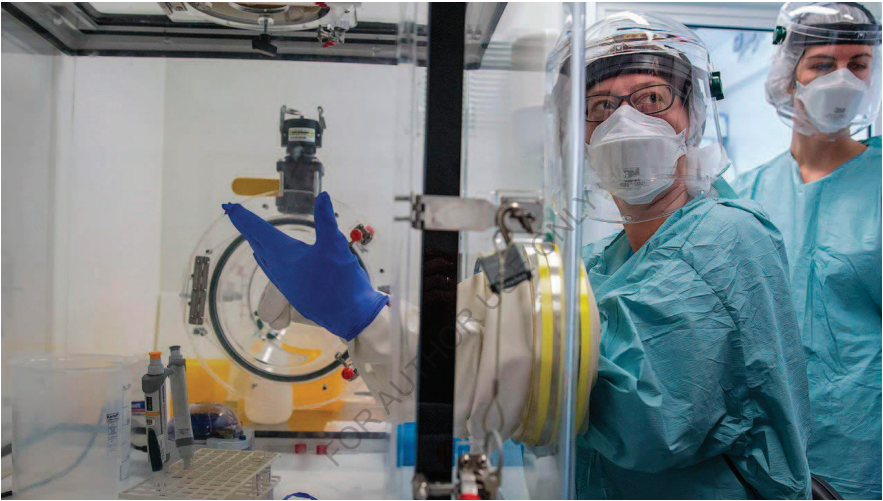
- Students must not be allowed to congregate in the halls outside the lab while waiting for practical exams to start. This is a fire hazard and disruptive to other classes. Give students a definite time to come to the lab to begin the practical.
- Personal Protective Equipment Safety glasses **MUST** be worn during ALL laboratory activity involving chemicals, boiling water, use of the Bunsen burner and dissecting by students, instructors, and laboratory aides. Other personal protective equipment (i.e. gloves) is to be worn as directed by laboratory setup instructions or the instructor. Failure to wear appropriate personal protective equipment in laboratory situations
- will lead to disciplinary measures, up to and including dismissal from employment or expulsion from the college program.
- Personnel working or using the lab must confine long hair, and wear regular length lab coats. When applicable, wear gloves to protect hands from contamination. Inspect gloves before use.
- NO clogs, flip-flops, open-toed shoes, sling back heels, sandals, or bare feet are allowed in lab. Regular length lab coats must be worn at all times over all clothing.
- Know the location of all fire and first aid equipment.

- Use of the Lab Students are NEVER allowed in the laboratory areas without supervision of the instructor or laboratory aide.
- Coats, purses, extra books, or any other materials not being used in the current laboratory activity are NOT allowed in the working area. These items are to be placed in the designated area during laboratory times.
- The doors to the laboratory and chemical storage areas MUST be locked upon leaving by the instructor and/or laboratory aide.
- Damaged or Defective Equipment All equipment and materials must be kept in its designated storage area when not in use. Inspect equipment and materials before use. NEVER used damaged or defective equipment.
- If equipment is believed to be damaged or defective: • Discard in the appropriate waste receptacle, OR • Remove from the work area and place a written notice on it of suspected damage or defect to prevent possible use, AND • Report damaged or defective equipment to the laboratory aide
- NO food, drink, smoking, or children are allowed in the college laboratory areas at ANY time.
- NEVER walk away from Bunsen burners when in use.

- BE ALERT for unusual or unsafe circumstances in the laboratory areas. REPORT all concerns IMMEDIATELY to the laboratory aide. If you make a mess, you are responsible for ensuring it is cleaned up.
- OSHA Laboratory Regulations Appendices to MIOSHA standard:
Hazardous Work in Laboratories Laboratory Safety System Audit Checklist Laboratory Safety System Procedures: Master Copy Instructor Handbook Laboratory Aide Handbook MSDS log Safety Inspection Checklist

FOR AUTHOR USE ONLY

SAFETY IN THE BIOLOGY LABORATORY



SAFETY IN THE BIOLOGY LABORATORY

GENERAL GUIDELINES

1. The safety rules are NOT flexible. All persons will follow all the rules all of the time.
2. All labs and activities are a team effort. Each lab group member must actively participate with their assigned group. The ENTIRE GROUP will be responsible for the lab materials used, as well as set up AND clean up! Breakage/failure to follow directions or rules will result in a loss of points or detention or both.
3. Know the location of all safety equipment, such as first aid equipment, eyewashes, fire blankets, and fire extinguisher.
4. Do NOT perform unauthorized experiments.
5. Keep your work area clean and free from unnecessary papers, books, and equipment.
6. Always read through the experiment before beginning. Note any hazards or cautions. Wear older clothing and closed-toe shoes on lab days. Use goggles when instructed by your teacher.
7. Never put anything in your mouth to taste or eat in the laboratory unless you are specifically told to do so. Never eat, drink, or cook food in/from any laboratory equipment.
8. Never leave any hot plates, gas burners or jets, or water faucets on for longer than necessary. Never leave any equipment unattended, especially open flames or chemicals.

9. Report all accidents or spills immediately to your teacher.
10. Handle all toxic and flammable substances with care and in well-ventilated areas.
11. Always use clean laboratory equipment. Always clean laboratory equipment after use.
12. Clean your work area at the conclusion of the experiment as your teacher directs. Clean all glassware carefully and thoroughly, do not leave any soap or oily residue on or in glassware.

FOR AUTHOR USE ONLY

SPECIFIC GUIDELINES



SPECIFIC GUIDELINES

1. Do NOT heat chemicals unless you are told specifically to do so. A chemical that is harmless while cool can be dangerous when hot... like, BOOM!!!

2. Do NOT heat a liquid in a closed container (like a stopper in a test tube). The expanding gases produced may blow the container apart, injuring people... causing burns, scars, or worse!

3. Do NOT attempt to dry out completely test tubes or other glassware by forcing paper towels into them. Simply let them air dry.

4. Never pour chemicals back into the STOCK container. Take only the amount you need, and pour any excess down the drain, or in the specified container.

5. Properly dispose of broken glassware. Never leave broken glassware or attempt to hide it by kicking it under something or sweeping it into the sink. Tell your teacher about broken glassware immediately. 6. Never connect wires to house current... duh!

7. Never position the mouth of the test tube towards yourself or anyone else while heating a substance in it. Remember... burn scars can be permanent.

8. Long hair must be tied back and kept out of the way of hot plate surfaces or Bunsen burners. Remember, hair burns better than paper and most hair sprays are extremely flammable.

9. You must remain at your designated lab table, except to obtain necessary lab equipment or materials.

10. Do NOT put anything in the sinks, such as trash, pieces of food, or any paper products.

11. Always use sponges or squeegees to clean up excess water at your lab tables. Do not leave puddles of water on your lab tables.

12. White cloth towels are for CLEAN HANDS ONLY. Paper towels are to be used for cleaning/drying lab materials and not lab tables, unless otherwise directed. Use PAPER TOWELS SPARINGLY!!

FOR AUTHOR USE ONLY

GUIDELINES FOR GENERAL LABORATORY SAFETY



GUIDELINES FOR GENERAL LABORATORY SAFETY

General Rules and Regulations

Each laboratory is unique in its operation and carries its own risk. It is the obligation of users to observe safety rules and regulations of each laboratory. Below are general safety guidelines applicable to most laboratories.

1. Proper procedures and safety precautions should be followed when experiments are performed.
2. Users are required to observe the necessary legislative ordinances when life specimens are involved in the experiment. Regulations should be followed in handling materials of potential biological and chemical hazards.
3. Fire exits should be cleared of obstacles. Make sure that fire-fighting equipment is available at convenient locations. Emergency response procedures, alarm systems and building evacuation routes should be posted.
4. Personal protection equipment should be available and used in the laboratory whenever necessary. Proper Eye Protectors must be worn as an absolute minimum in laboratories using hazardous chemicals, dangerous machinery, laser equipment or biological agents.
5. Be alert to unsafe conditions and actions, and bring them to the attention of your supervisor or the Departmental Health and Safety Officer so that remedial action can be made as soon as possible. There should be strong

communication link between Departmental Health and Safety Officers and the Administrative Support & Estates Management Office. Sufficient professional advice and support should be obtained from these channels. Caring about the health and safety of your fellow workers will be rewarding. Incidents or accidents happening to them may involve you as well.

6. No food and beverages or use of make-up are allowed in the laboratory. Smoking is strictly prohibited. Glassware or containers that had been used for laboratory operations should never be used for preparing or keeping food or drinks. Food storage is not permitted in laboratory freezers, refrigerators, ice chests, or cold rooms.

7. Be fully understood the potential hazards of equipment and substances, and take the appropriate precautions before using them. For those potentially hazardous substances or equipment that you are not familiar with, refer to the Material Safety Data Sheets (MSDSs), technical manuals or other information sources kept in your department, or consult your supervisor.

8. Follow disposal procedures for chemical wastes. Experimental apparatus may require traps or scrubbing devices to prevent the escape of toxic substances into the laboratory and the environment.

9. Make sure that all containers of hazardous substances (chemicals, biological agents or radioactive substances) are correctly and clearly labeled.

10. Accidents often happen when heating processes are left unattended. Experiments involving a heating process should be constantly monitored. The built-in temperature control of heating equipment such as hot plates

and heating mantles should not be overly relied upon. When a stable and even temperature is required, a heating bath is recommended. Water baths can be used up to about 80oC. Paraffin is suitable up to about 150oC but is still flammable at high temperatures. Silicone oils can be used for temperatures up to 300oC.

11. Warning signs should be posted to alert attentions in the work area when unusual hazards, such as radiation, laser operations, dangerous chemicals, biological hazards, or other special hazards exist.

12. Remain out of the area of a fire or personnel injury unless it is your responsibility to handle the emergency. Curious bystanders interfere with rescue by emergency personnel, and endanger themselves.

13. Never distract or startle other workers. Practical jokes or horseplay in the laboratory cannot be tolerated at any time.

14. Equipment should only be used for its designed purpose. Inappropriate choice of equipment, such as the use of ordinary glassware for heat-generating chemical reactions, has led to serious accidents before.

15. Place and handle your reaction apparatus carefully. Do not use excessive force. Plan the experiment so that it is not necessary to move it until the reaction is completed.

16. Confine long hair and loose clothing when in the laboratory. Shoes must be worn at all times in buildings where chemicals are stored or used. Sandals or open toe shoes are not acceptable.

17. No mouth pipetting in the laboratory.

18. Avoid exposure to gases, vapours and aerosols in the laboratory. Use appropriate safety equipment and work in a fume hood whenever such exposure is likely.

19. Always wash your hands and forearms with soap and water before leaving the laboratory area. Washing with solvents can remove the natural protective oil layer from the skin and can cause irritation and inflammation.

20. Deal with spills of all hazardous substances without delay according to detailed procedures in respective Departmental Health and Safety Guidelines. Potential leaks of any substances, including lubrication oil from vacuum pumps, should be dealt with by using a suitable drip tray as a precaution to contain the spills.

The following guidelines were designed by the Biology Department Faculty for use in all biology courses offered at Evergreen Valley College. It is mandatory that students follow these regulations. Students are expected to read these regulations, sign the Safety Form on page 5, and return the bottom half to the instructor. This procedure must be completed before a student can begin any laboratory activity. The student should keep the regulations for future reference.

GENERAL SAFTY RULES IN LABORATORY



GENERAL SAFTY RULES IN LABORATORY

1. Read all laboratory instructions before coming to class.
2. Dress appropriately for lab: closed-toe shoes, long hair tied back, no dangling jewelry or loose clothing. Avoid wearing contact lenses.
3. Know the location and proper use of the following in your laboratory: first aid kit, fire extinguisher, eye wash, emergency telephone, fire blanket, closest fire alarm, chemical shower, emergency exits, emergency gas shut-off valves, evacuation site for your classroom. Keep the safety equipment and exits free of obstruction.
4. Behave in a professional manner.
5. Only registered students are allowed in the laboratory.
6. Students are not permitted access to prep rooms, laboratory cupboards, or instructor desks without prior approval from the instructor.
7. Students who have any medical conditions (e.g. diabetes, immunological defect, pregnancy, epilepsy) that may necessitate special precautions in the laboratory must inform the professor.
8. To encourage classroom unity and promote safety, please speak English in class.
9. No food or drink (including water) allowed in the laboratory.
10. Keep hands away from face, eyes and clothing while using hazardous solutions or materials. Do not handle contact lenses or apply cosmetics during class.

11. Maintain a clean, open work area free of everything except material related to the exercise.
12. Use all equipment and materials with care and as authorized by instructor.
13. Report all spills or accidents to the instructor. Clean up spills immediately.
14. Wipe down work surfaces at the beginning and end of each lab.
15. Wash your hands at the beginning and end of class.
16. At the end of lab make sure all items are returned to their proper location. Push in chairs. Make sure sinks and your work space are clean and free of debris.

FOR AUTHOR USE ONLY

**EMERGENCY
PREPAREDNESS**



EMERGENCY PREPAREDNESS

1. During an earthquake stay in the room and immediately get under the desk. After the earthquake follow your instructor's directions.
2. During a fire or other emergencies requiring evacuation, meet at the designated evacuation location. Students are required to meet and remain at the evacuation site until dismissed by the instructor.
3. During a shelter-in-place emergency, students should remain in the classroom and follow instructions.

FOR AUTHOR USE ONLY

FOR AUTHOR USE ONLY

HANDLING EQUIPMENT AND MATERIALS



HANDLING EQUIPMENT AND MATERIALS

1. Properly label all glassware and slides.
2. Keep all equipment and materials away from the edge of the lab benches.
3. Use protective aids (e.g. test-tube clamps, hot pads) when handling hot glassware.
4. Light gas burners only as instructed. Be sure no volatile or flammable materials are being used nearby. Remember to have long hair tied back, no dangling jewelry or loose clothing
5. When heating a test tube do not look into the mouth of the tube or point it into the direction of any person.
6. Heat sources should only be on when in use. Do not leave heat sources unattended. When in doubt, students should assume a heat source is on or hot.
7. Operate electrical equipment in a dry area with dry hands.
8. When removing an electrical plug from its socket, pull the plug, not the cord.
9. Inform your instructor immediately if equipment or materials are broken or not working properly.
10. Never return reagents back into their original containers. Keep containers capped when not in use.

FOR AUTHOR USE ONLY

GENERAL SAFTY RULES IN LABORATORY

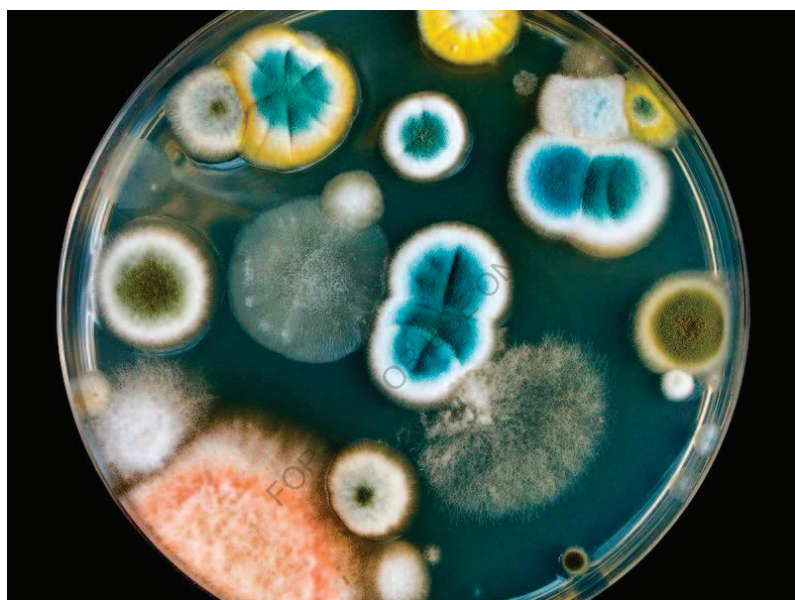


HANDLING HAZARDOUS MATERIAL

1. Wear appropriate eye protection as directed by the instructor.
2. Wear disposable gloves when handling hazardous materials including chemicals, organisms, or bodily fluids.
3. Wear protective clothing when required by the instructor.
4. Follow correct disposal procedures as detailed in “Cleanup and Disposal.”
5. To prevent contamination by needle stick injuries, always use new needles and lancets.
6. Use the fume hood whenever noxious, corrosive or toxic fumes are produced or released.

FOR AUTHOR USE ONLY

BACTERIA AND FUNGI



BACTERIA AND FUNGI

1. Never open Petri dishes containing bacterial or fungal growth unless directed to do so by the instructor.
2. Dispose of all discarded bacterial and fungal cultures as directed by the instructor.

FOR AUTHOR USE ONLY

FOR AUTHOR USE ONLY

CLEANUP AND DISPOSAL



CLEANUP AND DISPOSAL

1. Follow correct disposal procedures – dispose of used materials promptly and in the appropriate disposal containers.
 - a. Dispose of broken glass in labeled glass disposal containers.
 - b. Dispose of needles and lancets in puncture-proof sharp containers
 - c. Dispose of used chemicals in labeled and capped waste containers as directed by your instructor.
 - d. For disposal of items contaminated with organisms or bodily fluids,
 - i. Decontaminate slides, spectrophotometer tubes, and other glassware in 10% bleach
 - ii. Dispose of soft items (i.e. gauze or swabs) in labeled biohazard containers.
 - iii. Dispose of all animal tissues in labeled biohazard containers.
2. Place micropipette tips, sample vials, and contaminated glassware in a labeled waste container or designated area.
3. See instructor for disposal of cadaver tissues.

FOR AUTHOR USE ONLY

SAFETY PLEDGE



SAFETY PLEDGE

I have read the Biology Laboratory Safety Guidelines and understand the rules and procedures of proper safety in the biology laboratory course listed below. I recognize my responsibility and pledge to observe all safety rules in the biology laboratory at all times.

I acknowledge that the instructor has covered all safety guidelines and equipment that we may need to use this semester.

Name of Student (please print) _____

Signature of Student _____

Date _____

Course _____ Section _____ Professor _____

SAFETY PLEDGE

I have read the Biology Laboratory Safety Guidelines and understand the rules and procedures of proper safety in the biology laboratory course listed below. I recognize my responsibility and pledge to observe all safety rules in the biology laboratory at all times.

I acknowledge that the instructor has covered all safety guidelines and equipment that we may need to use this semester.

Name of Student (please print) _____

Signature of Student _____

Date _____

Course _____ Section _____ Professor _____

FOR AUTHOR USE ONLY

GLOSSARY



GLOSSARY

Aboral	: Opposite the mouth
Amoeboid	: Cell movements resembling those of the amoeba.
Angstrom	: One thousand of a micron.
Archenteron	: Primitive digestive tract of a metazoan embryo, formed during gastrulation.
Autotrophic	: Nutrition. Process of nutrition in which an organism manufactures its own food.
Basal disc	: Foot of some Cnidaria which is flattened and attaches to a substratum by secretion of a sticky substance.
Binary fission	: the type of asexual reproduction by means of which the organism divides into two Approximately equal halves.
Buccal	: Pertaining to the mouth or oral cavity.
Cnidaria or Coelenterata	: Phylum of animals all possessing cnidoblast structures.
Cnidoblast	: Type of cell in which nematocyst is found.
Coelom	: The body cavity lined with tissue of mesodermal origin in which the digestive and other organs lie.
Conjugation	: A method of sexual reproduction in which two unicellular animals untie, exchange nuclear material and then divide as in the Paramecium.
Contractive vacuole	: A space in the cytoplasm of certain species of protozoa where fluids collect before being periodically discharged to the outside.
Ctenophora	: Radiate phylum of animals possessing comb-such as comb-jellies.
Cuticle	: Thin non-cellular outermost secreted by the underlying epidermis.
Cyst	: The stage of an organism where it is enclosed in a resistant wall.
Cytopharynx	: Pharynx or gullet of a protozoan such as Paramecium.
Diploblastic	: Derived from two embryonic germ layers, ectoderm and endoderm.

Enteron	: Digestive tract, especially in Cnidaria.
Entoprocta	: Pseudocoelomate, sessile phylum with U-shaped intestine, mouth surrounded by circle of ciliated tentacles, and opening within circle.
Extracellular	: Outside of the cell or cells.
Exumbrella	: Convex, aboral surface of the medusa.
Fission	: Asexual method of reproduction by division into two or more approximately equal in size.
Food vacuole	: Intra – cellular digestive organelle.
Free-living	: Capable of independent existence.
Gastrodermis	: Lining of coelenterate digestive cavity.
Gastrulation	: Process by which two germ layers, ectoderm and endoderm.
Holophytic	: Type of nutrition, found in green plants and in some mastigophores, which involves photosynthesis.
Holozoic	: Type of nutrition found in most animals, that involves ingestion and digestion of organic material.
Hydrotheca	: Transparent membrane that extends from the perisarc and surrounds the main part of the hydranth.
Hypostome	: Region surrounding the mouth in coelenterates.
Inter-cellular	: Between cells.
Intra-cellular	: Within cells.
Isogamy	: Sexual reproduction involving fusion of two similar gametes but from opposite sexes.
Kinetosome	: The basal body of a flagellum or cilium.
Lophophore	: Anterior tentacle – bearing area of certain coelomates; serves in food capture.
Mesogloea	: Non-cellular jelly-like substance lying between the ectoderm and endoderm in coelenterates.
Metagenesis	: Alternation of sexual with an asexual generation in reproduction in the life cycle of a coelenterate such as Obelia.
Myoneme	: Type of contractile fibril in certain Protozoa.
Nephridiopore	: External opening of an excretory tubule or nephridium.

Pedal	: Pertaining to father.
Pellicle	: The protective layer on the surface of some protozoans, for example, Paramecium.
Penetrant	: Largest type of cnidarians nematocyst, containing a coiled tube and spines, used in prey capture.
Peristome	: Region around the mouth of a radially symmetrical animal such as hydra.
Phagocyte	: Type of white blood cell that engulfs and digests bacteria and other foreign materials.
Pinocytosis	: Cellular drinking or intake of fluid.
Plankton	: Floating or drifting aquatic organisms, mostly microscopic.
Plasmasol	: Relatively liquid cytoplasm.
Pneumatophore	: Air – filled float of siphonophoran hydroids.
Polyp	: A tubular coelenterate forms.
Prosopyle	: One of the surface pores opening into a sponge chamber.
Protozoa	: A phylum of cellular animals.
Proximal	: Near the point of attachment of an organ.
Pseudocoel	: A body cavity not completely lined with mesoderm as found in round worms.
Pseudopodia	: Blunt temporary protoplasmic projections found in ameba or in some ameba like cells.
Schizocoel	: The coelom formed by the splitting of embryonic mesoderm.
Sedentary	: Staying in one place.
Siliceous	: Containing silicon dioxide or silica.
Spicule	: One of many solid structures that composed the structural framework of a sponge.
Spongocoel	: Paragastric or central cavity of a sponge.
Syngamy	: Union of gametes in sexual reproduction forming a zygote.
Taxis	: A movement response

Tentacle	: A flexible arm like extension from the body of many invertebrates such as hydra.
Tentaculocyst	: Sense organs of some cnidarians.
Triplocyst	: Derived from three primary germ layers-ectoderm, mesoderm, and endoderm.
Vestibule	: An outer cavity with an entrance to a (usually) larger, deeper cavity.
Zoid	: One of the members of a hydroid or siphonophore colony.

FOR AUTHOR USE ONLY

REFERENCES



PHYLUM
The Annelids
Phylum of Life Sciences

ANNELIDA

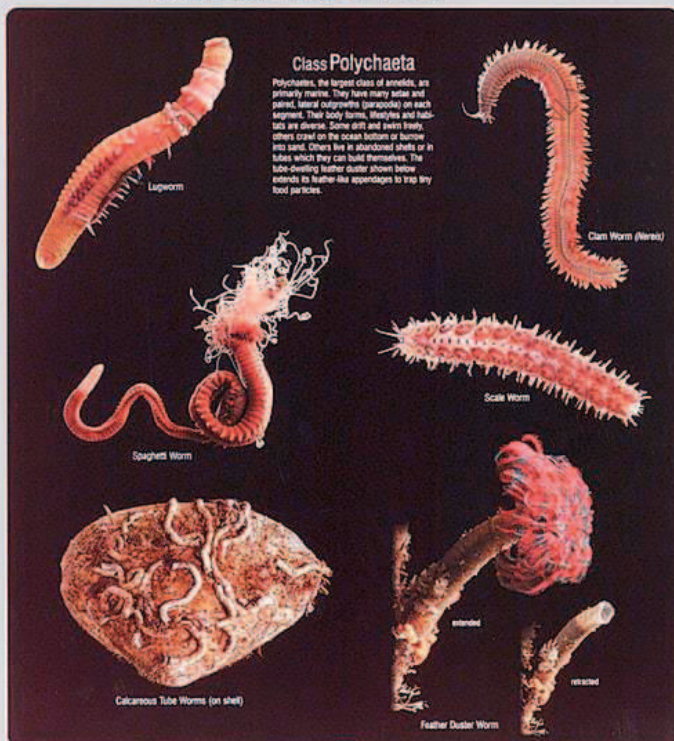
Annelids are found in almost all marine, freshwater and terrestrial habitats. They are the most highly developed worms with a true body cavity and a more complex circulatory system. Their most characteristic feature is the partitioning of the body into ring-like segments. Most annelids, other than leeches, have external bristles called setae on nearly all of the segments. Unless noted, the specimens on this chart are 1 to 2 times life size.

BioCam

Copyright © 1997

Printed in the USA

www.biocam.com



Class Hirudinea

Most leeches live in freshwater habitats and are scavengers or predators feeding on small invertebrates. Some also feed by attaching themselves with suckers to a host and drawing large quantities of blood. Their body is usually flattened and they lack parapodia and setae.



REFERENCES

- Barnes, R.D.** 1980 invertebrate zoology, W.B. Saunders Company, Philadelphia and London.
- Berril, N.J.** , 1957, Indestructible Hydra Scientific American, December, 1957.
- Berril, N.J.**, 1966, Biology in Action, Heinemann Educational Books Ltd., London, U.K.
- Chandler, A.C.**, and C.P. Read, 1961, I introduction to Parasitology, W.B. Saunders Company, Philadelphia and London.
- Cheng. T.C.** 1973, General Parasitology, Academic Press New York.
- Elliot, A.**, 1968, Zoology, Appleton – Century – crafts, Division of Meredith Corporation, New York, U.S.A.
- Gray, J.** and Lissman, H.W.1938, Studies of Animal Locomotion, VII. Locomotory reflexes in the Earthworm, J.Exp. Biol., 15 ; 506-517.
- Hall, R.P.**, 1953, Protozoology, Prentice Hall Inc. Englewood Cliffs, N.J., U.S.A.
- Hirschfield, H.I.**, 1962, The Biology of the Amoeba, Annals of the New York Academy of Sciences, Vol.78, Art 2, pp. 401-704.
- Hyman, L.H.**, 1940, TheInvertebraes, Protozoa through Ctenophora, Vol. I McGraw Hill Book Company, New York, U.S.A.
- Hyman, L.H.**, 1959. The Invertebraes, smaller Coelomate Groups, Vol. V, McGraw Hill Book Company, New york, U.S.A.
- Hyman L.H.**, 1967, The Invertebrates, Mollusca I, Vol. VI, McGraw Hill Book Company, New York, U.S.A.
- Imms, A.D.**, Richards, O.W., and Davies, R.G. 1957, A General Text Book of Entomology, Methuen and Company Ltd., London, U.K.
- Jardan, E.L.** and Verma P.S. Invertebrates Zoology S.Chand and Company. New Delhi. X Revised Edition.

Kotpal, R.L. Modern Text Book of Zoology, Invertebrates Animal Diversity-I, Rastogi Publications.

Manwell, R.D., 1961, Introduction to Protozoology, Edwin Arnold Publishers Ltd., London, U.K.

Mast, S.O., 1931, Locomotion in Amoeba Proteus, Protoplasma, 14: 321-330.

Mayr, E., 1963, Animal species and Evolution, Oxford University Press, New York.

Mercer, E.H.1959., An Electron Microscopic Study of Amoeba proteus. Proc.R.Soc. London B. 150: 216-232.

Parker, T.J. and William, A. Haswell edited by A.J., Marshall and W.D. Williams. (7th edition), 1972, A Text Book of Zoology: Invertebrates, English Language Book Society and Macmillan Company, London.

Russel-Hunter, W.D., 1968, A Biology of Lower Invertebrates, The Macmillan Co., New York.

Sedgewick, A., 1966, A Student Text Book of Zoology, I, III, Central Book Depot, Allahabad.

Sleigh, M., 1975, The Biology of Protozoa, Edwin Arnold (Publishers) Ltd. London. Storer, T.I., and R.L. Usinger, 1965, General Zoology, McGraw Hill Book company New York, London.

Verma, P.S., 1993, A Manual of Practical Zoology Invertebrates, S.Chand& Co. Ltd., New Delhi, India.

Verma, P.Srivastava, P.C 1998, Advanced Practical Zoology 3ed.

Vickerman, K and Cox, F.E.G., 1967, The Protozoa, John Murray, London.

Wilson, H.V., 1907, On some phenomena of coalescence and regeneration in sponges, J.Exp. Zool., 5: 245-257.

Wichterman, R., 1955, The Biology of Paramecium, The Blakiston Company, Inc. Toronto, New York, U.S.A.

FOR AUTHOR USE ONLY

FOR AUTHOR USE ONLY

**More
Books!**



yes
I want morebooks!

Buy your books fast and straightforward online - at one of the world's fastest growing online book stores! Environmentally sound due to Print-on-Demand technologies.

Buy your books online at
www.get-morebooks.com

Kaufen Sie Ihre Bücher schnell und unkompliziert online – auf einer der am schnellsten wachsenden Buchhandelsplattformen weltweit!
Dank Print-On-Demand umwelt- und ressourcenschonend produziert.

Bücher schneller online kaufen
www.morebooks.de

SIA OmniScriptum Publishing
Brivibas gatve 197
LV-103 9 Riga, Latvia
Telefax: +371 68620455

info@omniscryptum.com
www.omniscryptum.com

OMNI Scriptum



FOR AUTHOR USE ONLY